Grover's Algorithm

Assessment of quantum search

ANSWER BOOK

This book accompanies the separate *Question Paper* (80 questions, 3 hours, 240 marks). It contains

1. the **answer key** with the topic tested by every question (page 2), and
2. the **Answer Bank**: every question restated with the correct option marked ✓ and a complete solution with derivations (from page 4).

Marking scheme (every question)

Each question has **four** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

Full marks:	+3 if ONLY the correct option is chosen;
Zero marks:	0 if none of the options is chosen (question unanswered);
Negative marks:	−1 in all other cases (wrong option, or more than one option chosen).

How to evaluate

For each section count C = number of correct answers and W = number of questions answered wrongly (or with more than one bubble darkened). Section marks = $3C - W$; unanswered questions score 0. The total is out of 240. Every numerical answer was re-computed independently (NumPy, including gate-by-gate simulation of the Grover circuits) by `verify_answers.py` in the source folder.

Conventions (identical to the Question Paper)

- n qubits, $N = 2^n$ items, M marked items ($M = 1$ unless stated otherwise). A basis state is written $|x\rangle = |x_1 x_2 \ldots x_n\rangle$: **qubit 1 is the most significant bit** and is written first. (The accompanying notebook writes `N_QUBITS` for n and $M = 2^N$ for the number of items.)
- $|s\rangle = H^{\otimes n}|0\ldots0\rangle = \frac{1}{\sqrt{N}}\sum_x |x\rangle$. The oracle is $O|x\rangle = (-1)^{f(x)}|x\rangle$ with $f(x) = 1$ exactly on the marked items; the diffusion is $D = 2|s\rangle\langle s| - I$; one iteration is $G = DO$ (oracle first).
- $\sin\theta = \sqrt{M/N}$ with $0 < \theta \le \frac{\pi}{2}$; $|m\rangle$ and $|r\rangle$ are the uniform superpositions of the marked and the unmarked items. Then $G^k|s\rangle = \sin\big((2k+1)\theta\big)|m\rangle + \cos\big((2k+1)\theta\big)|r\rangle$, so $P_k = \sin^2\big((2k+1)\theta\big)$ and $k_{\text{opt}} = \text{round}\left(\frac{\pi}{4\theta} - \frac{1}{2}\right)$.
- **Textbook circuit:** $H^{\otimes n}$, then k times: [X on each qubit where x^* has a 0; a multi-controlled Z on all n qubits; the same X gates] followed by [$H^{\otimes n}$; $X^{\otimes n}$; a multi-controlled Z; $X^{\otimes n}$; $H^{\otimes n}$]; finally measure all qubits.
- $H = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix}1 & 1\\ 1 & -1\end{smallmatrix}\right)$; X, Y, Z are the Pauli matrices; $|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$. Two-qubit matrices use the basis order $|00\rangle, |01\rangle, |10\rangle, |11\rangle$.
- All measurements are in the computational basis unless stated otherwise; log means $\log_2$. A numerical value given in a question may be used without a calculator.

Answer key

Answer distribution: (A) 20, (B) 20, (C) 20, (D) 20.

Section A — Mathematics for Grover

Q.	Ans.	Topic tested	Q.	Ans.	Topic tested
1	**B**	Triple-angle identity	11	**A**	The signed sum $N - 2M$
2	**C**	Reflection matrix $2uu^T - I$	12	**B**	Double-angle formula
3	**D**	Two reflections make a rotation	13	**B**	Square-root scaling
4	**D**	Eigenvalues of a rotation	14	**C**	Period of the $\sin^2$ curve
5	**A**	Powers of a rotation	15	**C**	$\sin^2 3\theta$ as a polynomial
6	**D**	Optimal integer k; why $-\frac{1}{2}$ matters	16	**D**	Traces of projectors and reflections
7	**A**	Inversion about the mean	17	**C**	Rounding rule vs. $\arcsin x > x$
8	**C**	Classical search without replacement	18	**A**	Spectrum of a reflection
9	**A**	Repetition and the complement rule	19	**D**	First maximum and period of $p(k)$
10	**B**	Counting solutions of a Boolean function	20	**B**	Tolerance window around the peak

Section B — Quantum Mechanics

Q.	Ans.	Topic tested	Q.	Ans.	Topic tested
21	**B**	Uniform state, partial outcome	31	**A**	Overlap $\cos\theta$
22	**A**	Sign flip and Born rule	32	**D**	State-vector memory
23	**C**	Hadamard layer signs	33	**A**	Diffusion for one qubit
24	**C**	Phase kickback with $\|-\rangle$	34	**D**	$\|s\rangle$ is fixed by D
25	**D**	No kickback with $\|+\rangle$	35	**C**	Single-qubit gate actions
26	**C**	X conjugation selects the item	36	**B**	Product vs. entangled states
27	**D**	CCZ from Toffoli	37	**A**	Why the oracle alone reveals nothing
28	**A**	Global vs. relative phase	38	**C**	Relative phases are observable
29	**B**	Subspace measurement	39	**D**	Toffoli as a phase oracle
30	**B**	Unitary vs. projector	40	**B**	Ancilla stays unentangled

Section C — Grover Basics

Q.	Ans.	Topic tested	Q.	Ans.	Topic tested
41	**D**	$N = 4$ is exact in one step	51	**D**	$M = 3N/4$
42	**B**	One iteration, $N = 8$	52	**A**	Over-rotation at $2k_{\text{opt}}$
43	**D**	k_{opt} and P for $n = 4$	53	**A**	Size of the speed-up
44	**B**	k_{opt} for $n = 15$	54	**B**	Shot statistics
45	**C**	Building the oracle	55	**A**	k_{opt} against n
46	**A**	Gate count per iteration	56	**A**	Blocks of the circuit
47	**B**	Diffusion circuit $= -D$	57	**B**	Independence of the marked item
48	**C**	Unmarked amplitude	58	**D**	Oscillation and $\sqrt{N}$ scaling
49	**C**	$M = N/4$	59	**C**	Amplitude trace, first iteration
50	**D**	$M = N/2$	60	**C**	Amplitude trace, second iteration

Section D — Advanced Grover

Q.	Ans.	Topic tested	Q.	Ans.	Topic tested
61	**B**	Several solutions: k_{opt}	71	**B**	Simulation cost
62	**A**	BBHT threshold for unknown M	72	**C**	Eigenvectors of G
63	**D**	Exact Grover search	73	**D**	Fixed-point search
64	**D**	Optimality (BBBV, Zalka)	74	**C**	Repeat-until-success cost
65	**C**	Parallel search	75	**A**	Query complexities
66	**B**	Amplitude amplification cost	76	**B**	Variants of Grover
67	**D**	Eigenphases of G	77	**D**	Grover and NP-completeness
68	**A**	Key search	78	**A**	Unknown M; flat curve at $M = N/2$
69	**C**	Noise and depth	79	**B**	Quantum counting outcomes
70	**A**	Toffoli count	80	**C**	Quantum counting estimate

Quick-marking grid

1 **B**	2 **C**	3 **D**	4 **D**	5 **A**	6 **D**	7 **A**	8 **C**	9 **A**	10 **B**
11 **A**	12 **B**	13 **B**	14 **C**	15 **C**	16 **D**	17 **C**	18 **A**	19 **D**	20 **B**
21 **B**	22 **A**	23 **C**	24 **C**	25 **D**	26 **C**	27 **D**	28 **A**	29 **B**	30 **B**
31 **A**	32 **D**	33 **A**	34 **D**	35 **C**	36 **B**	37 **A**	38 **C**	39 **D**	40 **B**
41 **D**	42 **B**	43 **D**	44 **B**	45 **C**	46 **A**	47 **B**	48 **C**	49 **C**	50 **D**
51 **D**	52 **A**	53 **A**	54 **B**	55 **A**	56 **A**	57 **B**	58 **D**	59 **C**	60 **C**
61 **B**	62 **A**	63 **D**	64 **D**	65 **C**	66 **B**	67 **D**	68 **A**	69 **C**	70 **A**
71 **B**	72 **C**	73 **D**	74 **C**	75 **A**	76 **B**	77 **D**	78 **A**	79 **B**	80 **C**

Answer Bank — questions with detailed solutions

SECTION A · Mathematics for Grover Q.1–20

Q.1 SINGLE-CORRECT MCQ · Triple-angle identity

If $\sin\theta = \frac{1}{4}$, the value of $\sin^2 3\theta$ is

(A) $\frac{1}{16}$ **(B)** ✓ $\frac{121}{256}$ **(C)** $\frac{9}{16}$ **(D)** $\frac{11}{16}$

Answer: (B) **Detailed solution**

What you need to know. Triple-angle identity $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$. In Grover's algorithm $P_1 = \sin^2 3\theta$.

Step-by-step solution. Use the triple-angle identity $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$:

$$\sin 3\theta = \frac{3}{4} - 4\cdot\frac{1}{64} = \frac{12}{16} - \frac{1}{16} = \frac{11}{16}, \qquad \sin^2 3\theta = \frac{121}{256}.$$

This is exactly the success probability of one Grover iteration on $N = 16$ items. The value $\frac{9}{16} = (3\sin\theta)^2$ is the small-angle estimate; it ignores the cubic correction.

Why the other options are wrong.

(A) $\frac{1}{16} = \sin^2\theta$ is the probability before any iteration.

(C) $\frac{9}{16} = (3\sin\theta)^2$ uses the small-angle estimate $\sin 3\theta \approx 3\sin\theta$ and drops $-4\sin^3\theta$.

(D) $\frac{11}{16}$ is $\sin 3\theta$ itself: the final squaring was forgotten.

Remember. Keep the exact identity for small N; the small-angle estimate is only reliable for large N.

Q.2 SINGLE-CORRECT MCQ · Reflection matrix $2uu^T - I$

Let $u = (\frac{3}{5}, \frac{4}{5})^T$. The matrix $2uu^T - I$ equals

(A) $\frac{1}{25}\begin{pmatrix}18 & 24\\ 24 & 32\end{pmatrix}$ **(B)** $\frac{1}{25}\begin{pmatrix}-7 & 12\\ 12 & 7\end{pmatrix}$ **(C)** ✓ $\frac{1}{25}\begin{pmatrix}-7 & 24\\ 24 & 7\end{pmatrix}$ **(D)** $\frac{1}{25}\begin{pmatrix}7 & -24\\ -24 & -7\end{pmatrix}$

Answer: (C) **Detailed solution**

What you need to know. uu^T is the matrix with entries $u_i u_j$. $2uu^T - I$ reflects vectors in the line along u: it fixes u and negates the perpendicular direction.

Step-by-step solution. $uu^T = \frac{1}{25}\left(\begin{smallmatrix}9 & 12\\ 12 & 16\end{smallmatrix}\right)$, so $2uu^T = \frac{1}{25}\left(\begin{smallmatrix}18 & 24\\ 24 & 32\end{smallmatrix}\right)$. Subtracting $I = \frac{1}{25}\left(\begin{smallmatrix}25 & 0\\ 0 & 25\end{smallmatrix}\right)$ gives $\frac{1}{25}\left(\begin{smallmatrix}-7 & 24\\ 24 & 7\end{smallmatrix}\right)$. *Check:* it fixes u, since $\frac{1}{25}\cdot\frac{1}{5}(-21+96,\ 72+28) = \frac{1}{125}(75, 100) = (\frac{3}{5}, \frac{4}{5})$. Its trace is 0 and its determinant is -1, as for every reflection in the plane.

Why the other options are wrong.

(A) This is $2uu^T$ alone: the I was not subtracted, and the matrix is not orthogonal.

(B) The off-diagonal entries are those of uu^T and were not doubled; the columns are then not unit vectors.

(D) This is $I - 2uu^T$, the negative of the required matrix; it sends u to $-u$.

Remember. Check a reflection by testing that it fixes u, and that its trace is 0 and determinant -1 in the plane.

Q.3 SINGLE-CORRECT MCQ · Two reflections make a rotation

A vector in the plane is reflected first in the line through the origin at angle $10°$ to the x-axis, and then in the line through the origin at angle $35°$. The combined map is

(A) a reflection in the line at angle $45°$ **(B)** a rotation by $+25°$

(C) a rotation by $-50°$ **(D)** ✓ a rotation by $+50°$

Answer: (D) Detailed solution

What you need to know. Reflection in the line at angle α: $\phi \mapsto 2\alpha - \phi$. Two reflections compose to a rotation by twice the angle between the lines.

Step-by-step solution. A reflection in the line at angle α sends a vector at angle ϕ to angle $2\alpha - \phi$. Applying $\alpha = 10°$ and then $\beta = 35°$:

$$\phi \;\to\; 20° - \phi \;\to\; 70° - (20° - \phi) = \phi + 50°.$$

The result is a rotation by $2(\beta - \alpha) = +50°$: **twice** the angle between the lines, in the direction from the first line to the second. This is the geometric heart of Grover's algorithm.

Why the other options are wrong.

(A) Two reflections have determinant $(-1)(-1) = +1$, so the product is a rotation and cannot be a reflection.

(B) $25°$ is the angle between the lines; the rotation angle is twice this.

(C) $-50°$ is the result for the opposite order (first $35°$, then $10°$).

Remember. Two reflections make a rotation by twice the angle between their axes: this is why G rotates by 2θ.

Q.4 SINGLE-CORRECT MCQ · Eigenvalues of a rotation

Let $\sin\theta = \frac{1}{4}$ with $0 < \theta < \frac{\pi}{2}$, and let $G = \begin{pmatrix} \cos 2\theta & -\sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{pmatrix}$. The eigenvalues of G are

(A) $\dfrac{7 \pm \sqrt{15}}{8}$ **(B)** $\dfrac{1 \pm i\sqrt{15}}{4}$ **(C)** ± 1 **(D)** ✓ $\dfrac{7 \pm i\sqrt{15}}{8}$

Answer: (D) Detailed solution

What you need to know. A plane rotation by α has eigenvalues $e^{\pm i\alpha}$, and $\cos 2\theta = 1 - 2\sin^2\theta$.

Step-by-step solution. A rotation by angle α has eigenvalues $e^{\pm i\alpha} = \cos\alpha \pm i\sin\alpha$ (trace $2\cos\alpha$, determinant 1). Here $\alpha = 2\theta$, with

$$\cos 2\theta = 1 - 2\sin^2\theta = 1 - \frac{2}{16} = \frac{7}{8}, \qquad \sin 2\theta = 2 \cdot \frac{1}{4} \cdot \frac{\sqrt{15}}{4} = \frac{\sqrt{15}}{8}.$$

Therefore the eigenvalues are $\frac{7 \pm i\sqrt{15}}{8}$; their modulus is $\sqrt{(49 + 15)/64} = 1$, as required.

Why the other options are wrong.

(A) These real numbers are not of modulus 1; the factor i was lost.

(B) These are $e^{\pm i\theta}$, using θ instead of the rotation angle 2θ.

(C) ± 1 are the eigenvalues of a reflection, or of a rotation by 0 or π only.

Remember. Eigenvalues of a rotation lie on the unit circle at the rotation angle: quantum counting reads θ from them.

Q.5 SINGLE-CORRECT MCQ · Powers of a rotation

Let $R = \begin{pmatrix} \sqrt{3}/2 & -1/2 \\ 1/2 & \sqrt{3}/2 \end{pmatrix}$. The smallest positive integer k with $R^k = -I$ is

(A) ✓ 6 **(B)** 4 **(C)** 3 **(D)** 12

Answer: (A) Detailed solution

What you need to know. Powers of a rotation add angles: $R(\alpha)^k = R(k\alpha)$, and $R(180^\circ) = -I$.
Step-by-step solution. R is the rotation by 30° (since $\cos 30^\circ = \sqrt{3}/2$, $\sin 30^\circ = 1/2$), so R^k is the rotation by $30^\circ k$. $-I$ is the rotation by 180°, hence $30^\circ k = 180^\circ$ and $k = 6$. ($k = 12$ gives $R^{12} = +I$.)
Why the other options are wrong.
(B) $k = 4$ gives $R(120^\circ)$, whose entries are not 0 and ± 1.
(C) $k = 3$ gives $R(90^\circ)$, a quarter turn.
(D) $k = 12$ gives $R(360^\circ) = +I$, not $-I$.
Remember. Identify the rotation angle first; the power then needs only arithmetic.

Q.6 SINGLE-CORRECT MCQ · Optimal integer k; why $-\frac{1}{2}$ matters

Let $\sin\theta = \frac{1}{10}$, $0 < \theta < \frac{\pi}{2}$; numerically $\theta = 0.10017$. Among the integers $0 \le k \le 10$, the one that makes $\sin^2((2k+1)\theta)$ largest is

(A) 8 **(B)** 10 **(C)** 5 **(D)** ✓ 7

Answer: (D) Detailed solution

What you need to know. $\sin^2((2k+1)\theta)$ is largest when $(2k+1)\theta$ is closest to $\frac{\pi}{2}$, that is for k nearest to $\frac{\pi}{4\theta} - \frac{1}{2}$.
Step-by-step solution. We want $(2k+1)\theta$ as close as possible to $\frac{\pi}{2}$, i.e. k closest to $\frac{\pi}{4\theta} - \frac{1}{2}$. Here

$$\frac{\pi}{4\theta} - \frac{1}{2} = \frac{3.1416}{0.40067} - 0.5 = 7.841 - 0.5 = 7.341,$$

so $k = 7$: $15\theta = 1.5025$ rad is 0.068 below $\frac{\pi}{2}$, while $17\theta = 1.7028$ rad is 0.132 above it. Then $\sin^2 15\theta = 0.9953$ and $\sin^2 17\theta = 0.9827$. The tempting estimate $\frac{\pi}{4} \cdot 10 = 7.85$ rounds to 8, which is wrong: it omits the $-\frac{1}{2}$.
Why the other options are wrong.
(A) $k = 8$ comes from rounding $\frac{\pi}{4}\sqrt{N} = 7.85$; it overshoots, and $\sin^2 17\theta = 0.983 < 0.995$.
(B) $k = 10 = \sqrt{N}$ omits the factor $\frac{\pi}{4}$, giving $21\theta = 2.10$ rad and $\sin^2 \approx 0.74$.
(C) $k = 5$ stops too soon: $11\theta = 1.10$ rad and $\sin^2 \approx 0.79$.
Remember. Use $\frac{\pi}{4\theta} - \frac{1}{2}$, not $\frac{\pi}{4}\sqrt{N}$, when the exact integer matters.

Q.7 SINGLE-CORRECT MCQ · Inversion about the mean

The vector $v \in \mathbb{R}^8$ has seven entries equal to $+1$ and one entry equal to -1. Each entry v_j is replaced by $2\bar{v} - v_j$, where $\bar{v}$ is the mean of the entries. The new value of the entry that was -1 is

(A) ✓ $\frac{5}{2}$ **(B)** $\frac{3}{2}$ **(C)** $\frac{7}{4}$ **(D)** $\frac{1}{2}$

Answer: (A) **Detailed solution**

What you need to know. Inversion about the mean: $v_j \mapsto 2\bar{v} - v_j$. It preserves the length of v and leaves the mean unchanged.

Step-by-step solution. The mean is $\bar{v} = \frac{7-1}{8} = \frac{3}{4}$. The special entry becomes $2 \cdot \frac{3}{4} - (-1) = \frac{5}{2}$; each other entry becomes $\frac{3}{2} - 1 = \frac{1}{2}$. *Check:* the length is preserved, $7 \cdot \frac{1}{4} + \frac{25}{4} = 8 = 7 + 1$. Inversion about the mean reflects every entry through the mean, so an entry far below the mean ends up far above it.

Why the other options are wrong.

(B) $\frac{3}{2} = 2\bar{v}$ forgets the term $-v_j = +1$.

(C) $\frac{7}{4}$ comes from using the mean $\frac{7}{8}$ of a wrong sum and is not consistent with length preservation.

(D) $\frac{1}{2}$ is the new value of the *other* entries.

Remember. An entry a distance d below the mean ends a distance d above it. Always check that the length is preserved.

Q.8 SINGLE-CORRECT MCQ · Classical search without replacement

A box holds 31 cards of which exactly 3 are marked. Cards are drawn one at a time at random without replacement until the first marked card appears. The expected number of draws is

(A) $\frac{31}{4}$ **(B)** $\frac{31}{3}$ **(C)** ✓ 8 **(D)** 16

Answer: (C) **Detailed solution**

What you need to know. Without replacement, the expected position of the first of M marked cards among N is $\frac{N+1}{M+1}$ (symmetry of the gaps).

Step-by-step solution. The $M = 3$ marked cards cut the random order of the $N - M = 28$ unmarked cards into $M + 1 = 4$ gaps, and by symmetry each gap holds on average $\frac{28}{4} = 7$ unmarked cards. The first marked card therefore comes after 7 unmarked ones on average, on draw number 8. In general the expected number of draws is $\frac{N+1}{M+1} = \frac{32}{4} = 8$. The value $\frac{31}{3}$ would be right for drawing *with* replacement.

Why the other options are wrong.

(A) $\frac{31}{4} = \frac{N}{M+1}$ uses N where $N + 1$ is needed.

(B) $\frac{31}{3} = N/M$ is the expectation *with* replacement (geometric distribution).

(D) $16 = \frac{N+1}{2}$ is the formula for one marked card.

Remember. With $M = 1$ the formula gives the classical average $\frac{N+1}{2}$ that Grover is compared with.

Q.9 SINGLE-CORRECT MCQ · Repetition and the complement rule

One run of a procedure succeeds with probability $\frac{1}{4}$, independently of the other runs. The probability of at least one success in three runs is

(A) ✓ $\frac{37}{64}$ **(B)** $\frac{27}{64}$ **(C)** $\frac{3}{4}$ **(D)** $\frac{1}{64}$

Answer: (A) Detailed solution

What you need to know. Use the complement: $P(\text{at least one}) = 1 - (1-p)^r$ for independent runs.

Step-by-step solution. The complement is three failures: $(\frac{3}{4})^3 = \frac{27}{64}$. Hence $P = 1 - \frac{27}{64} = \frac{37}{64}$. Adding the probabilities ($3 \times \frac{1}{4} = \frac{3}{4}$) double-counts the runs in which more than one succeeds.

Why the other options are wrong.

(B) $\frac{27}{64}$ is the probability of three failures, i.e. the complement.

(C) $\frac{3}{4}$ adds probabilities of events that are not mutually exclusive.

(D) $\frac{1}{64}$ is the probability that all three runs succeed.

Remember. "At least one" questions are almost always solved by the complement.

Q.10 SINGLE-CORRECT MCQ · Counting solutions of a Boolean function

Let $f(x_1, x_2, x_3, x_4) = (x_1 \vee x_2) \wedge (x_3 \oplus x_4)$ on $\{0,1\}^4$. The fraction of inputs with $f = 1$ is

(A) $\frac{1}{2}$ **(B)** ✓ $\frac{3}{8}$ **(C)** $\frac{1}{4}$ **(D)** $\frac{5}{8}$

Answer: (B) Detailed solution

What you need to know. Count the solutions of a Boolean function. If factors share no variables, the counts multiply. $\sin^2\theta = M/N$.

Step-by-step solution. The two factors involve different variables, so the counts multiply. $x_1 \vee x_2 = 1$ for 3 of the 4 pairs (x_1, x_2); $x_3 \oplus x_4 = 1$ for 2 of the 4 pairs (x_3, x_4). So $M = 3 \times 2 = 6$ of $N = 16$ inputs, a fraction $\frac{6}{16} = \frac{3}{8}$. This fraction is the quantity $\sin^2\theta$ in a Grover search for the solutions of f.

Why the other options are wrong.

(A) $\frac{1}{2}$ is the fraction for $x_3 \oplus x_4$ alone.

(C) $\frac{1}{4} = \frac{4}{16}$ counts only two solutions of $x_1 \vee x_2$ instead of three.

(D) $\frac{5}{8}$ averages the fractions $\frac{3}{4}$ and $\frac{1}{2}$ instead of multiplying them.

Remember. Find M by factoring the Boolean expression into independent parts.

Q.11 SINGLE-CORRECT MCQ · The signed sum $N - 2M$

A Boolean function f on 4 bits takes the value 1 on exactly 3 inputs. The value of $\sum_{x \in \{0,1\}^4} (-1)^{f(x)}$ is

(A) ✓ 10 **(B)** 7 **(C)** 6 **(D)** 13

Answer: (A) Detailed solution

What you need to know. $(-1)^{f(x)}$ is $+1$ on the $N - M$ unmarked inputs and -1 on the M marked ones, so the sum is $N - 2M$.

Step-by-step solution. Each of the $N - M = 13$ inputs with $f = 0$ contributes $+1$, and each of

the $M = 3$ inputs with $f = 1$ contributes -1:

$$\sum_x (-1)^{f(x)} = (N - M) - M = N - 2M = 16 - 6 = 10.$$

Divided by N this is the factor $1 - \frac{2M}{N}$ that appears in the mean amplitude after the oracle.

Why the other options are wrong.

(B) 7 is $N/2 - 1$ and does not follow from the counts.

(C) $6 = 2M$ is the amount lost from N, not the sum.

(D) $13 = N - M$ counts the unmarked inputs and ignores the -1 from the marked ones.

Remember. The mean amplitude after the oracle is proportional to $N - 2M$.

Q.12 SINGLE-CORRECT MCQ · Double-angle formula

If $\sin^2\theta = \frac{1}{16}$, then $\cos 4\theta$ equals

(A) $\frac{1}{2}$ **(B)** ✓ $\frac{17}{32}$ **(C)** $\frac{3}{4}$ **(D)** $\frac{49}{64}$

Answer: (B) Detailed solution

What you need to know. $\cos 2\alpha = 1 - 2\sin^2\alpha = 2\cos^2\alpha - 1$. Apply it twice.

Step-by-step solution. Apply the double-angle formula twice. $\cos 2\theta = 1 - 2\sin^2\theta = 1 - \frac{1}{8} = \frac{7}{8}$, and then

$$\cos 4\theta = 2\cos^2 2\theta - 1 = 2 \cdot \frac{49}{64} - 1 = \frac{98 - 64}{64} = \frac{34}{64} = \frac{17}{32}.$$

$\frac{49}{64} = \cos^2 2\theta$ is an unfinished step, not the answer.

Why the other options are wrong.

(A) $\frac{1}{2}$ does not follow from the formula: it would mean $4\theta = 60°$.

(C) $\frac{3}{4}$ is $1 - 4\sin^2\theta$, a linear estimate valid only for small angles.

(D) $\frac{49}{64} = \cos^2 2\theta$ is an intermediate value.

Remember. Grover amplitudes are polynomials in $\sin\theta$, so double- and triple-angle formulas appear everywhere.

Q.13 SINGLE-CORRECT MCQ · Square-root scaling

Let $g(n) = \frac{\pi}{4}\sqrt{2^n}$. Then $g(n+6)/g(n)$ equals

(A) 6 **(B)** ✓ 8 **(C)** $\sqrt{6}$ **(D)** 64

Answer: (B) Detailed solution

What you need to know. $\sqrt{2^n} = 2^{n/2}$, so adding a bits multiplies $\sqrt{N}$ by $2^{a/2}$.

Step-by-step solution. $\frac{g(n+6)}{g(n)} = \sqrt{\frac{2^{n+6}}{2^n}} = \sqrt{2^6} = 2^3 = 8$. Six extra bits make the search space 64 times larger but the number of Grover iterations only 8 times larger.

Why the other options are wrong.

(A) 6 treats the cost as linear in the number of bits.

(C) $\sqrt{6}$ takes the root of the *number of added bits* rather than of 2^6.

(D) $64 = 2^6$ is the growth of N, which is the classical scaling.

Remember. Each extra qubit multiplies the iteration count by $\sqrt{2}$.

Q.14 Single-correct MCQ · Period of the $\sin^2$ curve

Let $\theta = \pi/10$. The smallest positive integer p such that $\sin^2\left((2k+2p+1)\theta\right) = \sin^2\left((2k+1)\theta\right)$ for every integer k is

(A) 10 **(B)** 20 **(C)** ✓ 5 **(D)** 2

Answer: (C) **Detailed solution**

What you need to know. Period of $\sin^2$ is π. Moving k by p adds $2p\theta$, so we need $2p\theta \in \pi\mathbb{Z}$.
Step-by-step solution. $\sin^2$ has period π. Replacing k by $k+p$ adds $2p\theta$ to the angle, so we need $2p\theta$ to be a multiple of π: $\frac{2p\pi}{10} = m\pi$, i.e. $p = 5m$. The smallest is $p = 5$. In general the curve repeats every $\frac{\pi}{2\theta}$ iterations when this is an integer. ($p = 10$ corresponds to the period 2π of sin itself, not to that of $\sin^2$.)
Why the other options are wrong.
(A) $p = 10$ is a period but not the smallest: it corresponds to 2π, the period of sin.
(B) $p = 20$ is also a period but not the smallest.
(D) $p = 2$ adds only $\frac{2\pi}{5}$ to the angle, not a multiple of π.
Remember. The success curve repeats every $\frac{\pi}{2\theta}$ iterations, so over-rotation is periodic.

Q.15 Matching lists · $\sin^2 3\theta$ as a polynomial

Match the value of $\sin^2\theta$ (List-I) with the value of $\sin^2 3\theta$ (List-II).

List-I		List-II	
(P)	$\sin^2\theta = \frac{1}{8}$	**(1)**	1
(Q)	$\sin^2\theta = \frac{1}{4}$	**(2)**	0
(R)	$\sin^2\theta = \frac{1}{2}$	**(3)**	$\frac{25}{32}$
(S)	$\sin^2\theta = \frac{3}{4}$	**(4)**	$\frac{1}{2}$

(A) P→4, Q→1, R→3, S→2 **(B)** P→3, Q→1, R→2, S→4
(C) ✓ P→3, Q→1, R→4, S→2 **(D)** P→1, Q→3, R→4, S→2

Answer: (C) **Detailed solution**

What you need to know. $\sin^2 3\theta = s(3-4s)^2$ with $s = \sin^2\theta$.
Step-by-step solution. Since $\sin 3\theta = \sin\theta\,(3-4\sin^2\theta)$, with $s = \sin^2\theta$ we get $\sin^2 3\theta = s(3-4s)^2$. $s = \frac{1}{8}$: $\frac{1}{8}\cdot\frac{25}{4} = \frac{25}{32}$ (3). $s = \frac{1}{4}$: $\frac{1}{4}\cdot 4 = 1$ (1). $s = \frac{1}{2}$: $\frac{1}{2}\cdot 1 = \frac{1}{2}$ (4). $s = \frac{3}{4}$: $\frac{3}{4}\cdot 0 = 0$ (2). Hence P→3, Q→1, R→4, S→2.
Why the other options are wrong.
(A) This gives $s = \frac{1}{8}$ the value $\frac{1}{2}$, but $\frac{1}{8}\cdot\frac{25}{4} = \frac{25}{32}$.
(B) This exchanges R and S; for $s = \frac{3}{4}$ we have $3 - 4s = 0$, so the value is 0.
(D) This exchanges P and Q; for $s = \frac{1}{4}$ the value is 1, not $\frac{25}{32}$.
Remember. $s = \frac{1}{4}$ is perfect, $s = \frac{1}{2}$ is flat and $s = \frac{3}{4}$ is the worst case for one iteration.

Q.16 MATCHING LISTS · Traces of projectors and reflections

Let $|u\rangle$ be a unit vector in $\mathbb{C}^N$, with $N \geq 3$. Match each matrix in List-I with its trace in List-II.

List-I		List-II	
(P)	$2\,|u\rangle\langle u| - I$	**(1)**	N
(Q)	$I - 2\,|u\rangle\langle u|$	**(2)**	$N-2$
(R)	$|u\rangle\langle u|$	**(3)**	1
(S)	I	**(4)**	$2-N$

(A) P→3, Q→2, R→4, S→1
(B) P→4, Q→2, R→1, S→3
(C) P→2, Q→4, R→3, S→1
(D) ✓ P→4, Q→2, R→3, S→1

Answer: (D) Detailed solution

What you need to know. $\operatorname{tr}|u\rangle\langle u| = \langle u|u\rangle = 1$ and trace is linear.
Step-by-step solution. The key fact is $\operatorname{tr}|u\rangle\langle u| = \langle u|u\rangle = 1$, and $\operatorname{tr} I = N$. So $\operatorname{tr}(2\,|u\rangle\langle u| - I) = 2 - N$ (4); $\operatorname{tr}(I - 2\,|u\rangle\langle u|) = N - 2$ (2); $\operatorname{tr}|u\rangle\langle u| = 1$ (3); $\operatorname{tr} I = N$ (1). Hence P→4, Q→2, R→3, S→1.
Why the other options are wrong.
(A) This gives $2\,|u\rangle\langle u| - I$ trace 1 and the projector trace $2 - N$, which ignores linearity.
(B) This swaps R and S: the projector has trace 1, the identity has trace N.
(C) This swaps P and Q: the sign in front of the identity decides which of $N-2$, $2-N$ applies.
Remember. Trace identifies eigenvalue multiplicities: D has one $+1$ and $N-1$ eigenvalues -1.

Q.17 ASSERTION–REASON · Rounding rule vs. $\arcsin x > x$

Assertion (A): Rounding $\frac{\pi}{4}\sqrt{N}$ to the nearest integer does not always give the integer k for which $(2k+1)\theta$ is closest to $\frac{\pi}{2}$, where $\sin\theta = 1/\sqrt{N}$.
Reason (R): For $0 < x < 1$ we have $\arcsin x < x$.

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A
(B) Both A and R are true, but R is *not* the correct explanation of A
(C) ✓ A is true, but R is false
(D) A is false, but R is true

Answer: (C) Detailed solution

What you need to know. Decide A and R independently. $\arcsin x = x + x^3/6 + \ldots$, so it exceeds x for $x > 0$.
Step-by-step solution. **A is true.** For $N = 100$, $\frac{\pi}{4}\sqrt{N} = 7.85$ rounds to 8, but the true optimum is $k = 7$ (see Q.6). **R is false:** the inequality goes the other way. $\arcsin x = x + \frac{x^3}{6} + \cdots > x$, for example $\arcsin 0.1 = 0.10017$. The actual reason for A is the omitted $-\frac{1}{2}$ in $k = \frac{\pi}{4\theta} - \frac{1}{2}$. Answer: (C).
Why the other options are wrong.
(A) R is false, so it cannot explain A.
(B) This needs R to be true, but the inequality in R points the wrong way.
(D) A is true: $N = 100$ is a counterexample to the rounding rule, and R is false.
Remember. An Assertion can be true while the Reason offered for it is false: test both with a number.

Q.18 Statement I / II · Spectrum of a reflection

Let $|u\rangle$ be a unit vector in $\mathbb{C}^N$, with $N \geq 2$, and let $D = 2\,|u\rangle\langle u| - I$.

Statement I: $D^2 = I$.

Statement II: D has the eigenvalue $+1$ with multiplicity 1 and the eigenvalue -1 with multiplicity $N-1$.

(A) ✓ Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true, but Statement II is false

(D) Statement I is false, but Statement II is true

Answer: (A) Detailed solution

What you need to know. D is a reflection: $D^2 = I$, it fixes $|u\rangle$, and it negates the orthogonal complement.

Step-by-step solution. **I is true:** $D^2 = 4\,|u\rangle\,\langle u|u\rangle\,\langle u| - 4\,|u\rangle\langle u| + I = I$, because $\langle u|u\rangle = 1$. **II is true:** $D\,|u\rangle = 2\,|u\rangle - |u\rangle = |u\rangle$, and for any $|w\rangle$ orthogonal to $|u\rangle$ we have $D\,|w\rangle = -\,|w\rangle$; such vectors span an $(N-1)$-dimensional space. *Check:* the trace is $1-(N-1) = 2-N$, in agreement with Q.16. Answer: (A).

Why the other options are wrong.

(B) Both are true: $\langle u|u\rangle = 1$ gives I, and acting on $|u\rangle$ and on its complement gives II.

(C) II is also true, as the trace check $1-(N-1) = 2-N$ confirms.

(D) I is true by direct expansion.

Remember. Every reflection is its own inverse and has only eigenvalues ± 1.

Paragraph for Questions 19 and 20

Define $p(k) = \sin^2\left((2k+1)\theta\right)$ for integers $k \geq 0$, with $\theta = \pi/18 = 10°$. (This is the shape of the Grover success curve, but the two questions below need only trigonometry.) Note that $(2k+1)\theta$ takes the values $10°, 30°, 50°, \ldots$

Q.19 Paragraph based · First maximum and period of $p(k)$

The smallest k with $p(k) = 1$, and the smallest positive period of the sequence $p(k)$, are

(A) first maximum at $k = 4$; smallest period 18

(B) first maximum at $k = 5$; smallest period 9

(C) first maximum at $k = 9$; smallest period 18

(D) ✓ first maximum at $k = 4$; smallest period 9

Answer: (D) Detailed solution

What you need to know. $p(k) = 1$ requires an odd multiple of $90°$. The period comes from 2θ steps compared with $180°$.

Step-by-step solution. $p(k) = 1$ requires $(2k+1)\cdot 10° = 90°$ (the first odd multiple of $90°$), i.e. $2k+1 = 9$, so $k = 4$. Moving from k to $k+q$ adds $20°q$ to the angle. $\sin^2$ has period $180°$, so we need $20°q = 180°m$; the smallest solution is $q = 9$. Hence period 9 ($q = 18$ corresponds to the period $360°$ of sin and is not the smallest).

Why the other options are wrong.

(A) 18 is a period but not the smallest: it corresponds to $360°$.

(B) $k = 5$ gives $110°$, already past the maximum.

(C) $k = 9$ gives 190°, which is near a minimum, and 18 is not the smallest period.

Remember. Convert to degrees and look at the list of angles: the sequence becomes easy to read.

Q.20 PARAGRAPH BASED · Tolerance window around the peak

The number of integers k with $0 \le k \le 17$ for which $p(k) \ge \frac{3}{4}$ is

(A) 3 **(B)** ✓ 6 **(C)** 9 **(D)** 4

Answer: (B) Detailed solution

What you need to know. $\sin^2\alpha \ge \frac{3}{4} \iff |\sin\alpha| \ge \frac{\sqrt{3}}{2}$; α is within 30° of an odd multiple of 90°.

Step-by-step solution. $\sin^2\alpha \ge \frac{3}{4}$ means $|\sin\alpha| \ge \frac{\sqrt{3}}{2}$, i.e. α lies within 30° of 90° or of 270°. For $k = 0, \ldots, 17$ the angle $(2k+1)\cdot 10°$ runs over $10°, 30°, \ldots, 350°$. The values in $[60°, 120°]$ are $70°, 90°, 110°$ ($k = 3, 4, 5$); the values in $[240°, 300°]$ are $250°, 270°, 290°$ ($k = 12, 13, 14$). That makes 6 values of k.

Why the other options are wrong.

(A) 3 counts only one of the two peaks in the range $0 \le k \le 17$.

(C) 9 is the period and not a count of angles in the window.

(D) 4 counts two values at each peak; each peak has three: $70°, 90°, 110°$ and $250°, 270°, 290°$.

Remember. A window of $\pm 30°$ around the peak corresponds to $P \ge \frac{3}{4}$: there is some tolerance around k_{opt}.

SECTION B · Quantum Mechanics Q.21–40

Q.21 Single-correct MCQ · Uniform state, partial outcome

The state $H^{\otimes 3}|000\rangle$ is measured in the computational basis. The probability that qubit 1 gives 1 and qubit 3 gives 0 is

(A) $\frac{1}{8}$ **(B)** ✓ $\frac{1}{4}$ **(C)** $\frac{1}{\sqrt{8}}$ **(D)** $\frac{1}{2}$

Answer: (B) Detailed solution

What you need to know. $H^{\otimes n}|0\ldots0\rangle$ is uniform and a product state, so its qubits are independent.

Step-by-step solution. All 8 outcomes have probability $\left|\frac{1}{\sqrt{8}}\right|^2 = \frac{1}{8}$. The event "$x_1 = 1$, $x_3 = 0$" contains the two outcomes $|100\rangle$ and $|110\rangle$, so $P = \frac{2}{8} = \frac{1}{4}$. (Equivalently the qubits are independent: $\frac{1}{2} \cdot \frac{1}{2}$.)

Why the other options are wrong.

(A) $\frac{1}{8}$ is the probability of one specific full string; here qubit 2 is free.

(C) $\frac{1}{\sqrt{8}}$ is an amplitude, not a probability.

(D) $\frac{1}{2}$ is the probability for one qubit only.

Remember. Sum over all full strings consistent with a partial outcome.

Q.22 Single-correct MCQ · Sign flip and Born rule

The state $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle)$ is measured in the computational basis. The probability of $|11\rangle$ is

(A) ✓ $\frac{1}{4}$ **(B)** 0 **(C)** $\frac{1}{2}$ **(D)** $-\frac{1}{4}$

Answer: (A) Detailed solution

What you need to know. Born rule: probability is |amplitude|^2, which ignores the sign.

Step-by-step solution. Born rule: $P = \left|-\frac{1}{2}\right|^2 = \frac{1}{4}$. The minus sign has no effect on this measurement. That is why the oracle step alone does not find the marked item: a second step (diffusion) must turn the sign into a change of magnitude.

Why the other options are wrong.

(B) 0 would mean the sign flip cancelled the amplitude, but it only changes its sign.

(C) $\frac{1}{2}$ is the modulus of the amplitude, not its square.

(D) Probabilities cannot be negative.

Remember. The oracle changes signs only; it does not change any measurement probability by itself.

Q.23 Single-correct MCQ · Hadamard layer signs

$H \otimes H$ is applied to $\frac{1}{2}(\,|00\rangle + |01\rangle + |10\rangle - |11\rangle\,)$. The amplitude of $|11\rangle$ in the result is

(A) $\frac{1}{2}$ **(B)** 0 **(C)** ✓ $-\frac{1}{2}$ **(D)** 1

Answer: (C) Detailed solution

What you need to know. $H^{\otimes 2}$ has entries $\frac{1}{2}(-1)^{x\cdot y}$; each output amplitude is a signed sum of the input amplitudes.

Step-by-step solution. Use $H^{\otimes 2}|x\rangle = \frac{1}{2}\sum_y (-1)^{x\cdot y}|y\rangle$. For $y = 11$ the signs $(-1)^{x\cdot y}$ for $x = 00, 01, 10, 11$ are $+, -, -, +$, so the amplitude is

$$\frac{1}{2}\cdot\frac{1}{2}[(+1)(1)+(-1)(1)+(-1)(1)+(+1)(-1)] = \frac{1}{4}(1-1-1-1) = -\frac{1}{2}.$$

The full result is $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle)$: this state is unchanged by $H \otimes H$.

Why the other options are wrong.

(A) $\frac{1}{2}$ has the wrong sign, from treating the entry for $x = 11$, $y = 11$ as negative.

(B) 0 would need the signed sum to cancel; here it is -2 before the factor $\frac{1}{4}$.

(D) 1 is the magnitude when all four terms align, which they do not.

Remember. Compute one output amplitude at a time with $(-1)^{x\cdot y}$.

Q.24 Single-correct MCQ · Phase kickback with $|-\rangle$

Let $U_f|x\rangle|y\rangle = |x\rangle|y \oplus f(x)\rangle$ and $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$. Then $U_f|x\rangle|-\rangle$ equals

(A) $|x\rangle|f(x)\rangle$ **(B)** $|x\rangle|-\rangle$ **(C)** ✓ $(-1)^{f(x)}|x\rangle|-\rangle$ **(D)** $|f(x)\rangle|-\rangle$

Answer: (C) Detailed solution

What you need to know. Phase kickback: an ancilla in $|-\rangle$ is an eigenvector of X with eigenvalue -1.

Step-by-step solution. Expand the ancilla: $U_f|x\rangle|-\rangle = \frac{1}{\sqrt{2}}|x\rangle\,(\,|f(x)\rangle - |1 \oplus f(x)\rangle\,)$. If $f(x) = 0$ the bracket is $|0\rangle - |1\rangle$; if $f(x) = 1$ it is $|1\rangle - |0\rangle = -(|0\rangle - |1\rangle)$. So the result is $(-1)^{f(x)}|x\rangle|-\rangle$. This is *phase kickback*: the ancilla is unchanged and the sign lands on the input register.

Why the other options are wrong.

(A) The ancilla does not store $f(x)$ here; that would be the case for an ancilla in $|0\rangle$.

(B) This drops the sign, which is the entire point of the construction.

(D) The input register is not overwritten; it remains $|x\rangle$.

Remember. Ancilla in $|-\rangle$ turns the bit oracle into the phase oracle.

Q.25 Single-correct MCQ · No kickback with $|+\rangle$

With U_f as in the previous question and $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, the state $U_f |x\rangle |+\rangle$ equals

(A) $|x\rangle |-\rangle$ **(B)** $(-1)^{f(x)} |x\rangle |-\rangle$ **(C)** $(-1)^{f(x)} |x\rangle |+\rangle$ **(D)** ✓ $|x\rangle |+\rangle$

Answer: (D) Detailed solution

What you need to know. $|+\rangle$ is an eigenvector of X with eigenvalue $+1$, so a controlled bit flip leaves it unchanged.

Step-by-step solution. $U_f |x\rangle |+\rangle = \frac{1}{\sqrt{2}} |x\rangle \left(|f(x)\rangle + |1 \oplus f(x)\rangle \right) = |x\rangle |+\rangle$ in both cases, because $|0\rangle + |1\rangle$ is symmetric under the bit flip ($X |+\rangle = + |+\rangle$). No phase is kicked back, so the ancilla must be prepared in $|-\rangle$, not $|+\rangle$.

Why the other options are wrong.

(A) U_f cannot change the ancilla from $|+\rangle$ to $|-\rangle$: flipping $|0\rangle + |1\rangle$ gives the same state.

(B) This is the result for an ancilla starting in $|-\rangle$.

(C) The sign requires the eigenvalue -1, that is the ancilla $|-\rangle$.

Remember. The ancilla's eigenvalue under X decides the kicked-back sign.

Q.26 Single-correct MCQ · X conjugation selects the item

Let CCZ be the three-qubit gate that multiplies only $|111\rangle$ by -1. Which basis state has its sign flipped by $X_2 \, \mathrm{CCZ} \, X_2$, where X_2 is X on qubit 2?

(A) $|111\rangle$ **(B)** $|010\rangle$ **(C)** ✓ $|101\rangle$ **(D)** $|000\rangle$

Answer: (C) Detailed solution

What you need to know. Conjugating by X moves the affected basis state: $X_2 \, \mathrm{CCZ} \, X_2 = I - 2 |101\rangle\langle 101|$.

Step-by-step solution. The rightmost X_2 acts first. A state picks up the sign exactly when X_2 maps it to $|111\rangle$, which happens for $|101\rangle$. The last X_2 restores the bit. So the sign flip sits on $|101\rangle$. This is the standard oracle recipe: apply X wherever the marked string has a 0.

Why the other options are wrong.

(A) $|111\rangle$ is the state for CCZ alone; after X_2 it is mapped to $|101\rangle$, which receives no sign.

(B) $|010\rangle$ is the bit-complement of the right answer.

(D) $|000\rangle$ would need X on all three qubits.

Remember. Put an X on every qubit where the marked string has a 0.

Q.27 Single-correct MCQ · CCZ from Toffoli

Let CCX be the Toffoli gate with target qubit 3. The product $(I \otimes I \otimes H) \, \mathrm{CCX} \, (I \otimes I \otimes H)$ equals

(A) CCX **(B)** $I \otimes I \otimes I$ **(C)** $Z \otimes Z \otimes Z$ **(D)** ✓ CCZ

Answer: (D) Detailed solution

What you need to know. $HXH = Z$ and $HZH = X$. Conjugating the target of a controlled gate conjugates the gate itself.

Step-by-step solution. Use $HXH = Z$ and $HIH = I$. In the branch where both controls are 1, the target sees $HXH = Z$; in all other branches it sees $HIH = I$. So the product is the doubly controlled Z, i.e. $\mathrm{CCZ} = \mathrm{diag}(1, \ldots, 1, -1)$.

Why the other options are wrong.

(A) CCX is the gate before conjugation; $HXH \neq X$.

(B) The product cannot be I because in the branch $|11\rangle$ the target sees Z.

(C) $Z \otimes Z \otimes Z$ flips the sign for every odd-weight string, not only for $|111\rangle$.

Remember. A multi-controlled Z is a multi-controlled X with Hadamards around the target.

Q.28 SINGLE-CORRECT MCQ · Global vs. relative phase

Which pair of states are physically *indistinguishable* (they give the same statistics for every possible measurement)?

(A) ✓ $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $-\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$

(B) $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ and $\frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$

(C) $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$

(D) $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $\frac{1}{\sqrt{2}}(|0\rangle + i\,|1\rangle)$

Answer: (A) Detailed solution

What you need to know. Global phase: $e^{i\gamma}|\psi\rangle$ is the same physical state. A relative phase is observable.

Step-by-step solution. Two state vectors describe the same physical state iff they differ by one overall factor $e^{i\gamma}$. Only the first pair does: the factor is -1. In each of the other pairs the *relative* phase differs, and a suitable measurement separates them. For example H maps $|+\rangle$ and $|-\rangle$ to $|0\rangle$ and $|1\rangle$. The minus sign in the diffusion circuit (see Q.47) is of the harmless global kind.

Why the other options are wrong.

(B) These two Bell states are orthogonal, so a suitable measurement separates them.

(C) $|+\rangle$ and $|-\rangle$ are orthogonal, and an X-basis measurement separates them perfectly.

(D) These differ by a relative phase i and have different $\langle Y\rangle$.

Remember. Test with one common factor across all components: if none exists, the states differ.

Q.29 SINGLE-CORRECT MCQ · Subspace measurement

Let $|\psi\rangle = \frac{1}{\sqrt{10}}(\,|00\rangle + 2\,|01\rangle - 2\,|10\rangle + i\,|11\rangle\,)$. A measurement asks only whether the two qubits have *different* values (outcome "different" on the span of $|01\rangle, |10\rangle$). The probability of "different" and the state afterwards are

(A) $\frac{4}{5}$ and $|01\rangle$

(B) ✓ $\frac{4}{5}$ and $\frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$

(C) $\frac{4}{5}$ and $\frac{1}{\sqrt{10}}(2\,|01\rangle - 2\,|10\rangle)$

(D) $\frac{2}{5}$ and $\frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$

Answer: (B) Detailed solution

What you need to know. Projective measurement: sum the probabilities in the subspace, project, and re-normalise.

Step-by-step solution. Sum the probabilities in the subspace: $P = \frac{4+4}{10} = \frac{4}{5}$. The post-measurement state is the projected vector, re-normalised: $\frac{(2|01\rangle - 2|10\rangle)/\sqrt{10}}{\sqrt{4/5}} = \frac{2|01\rangle - 2|10\rangle}{\sqrt{8}} = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$. Relative phases inside the subspace survive, and the measurement does not pick one of the basis states.

Why the other options are wrong.

(A) The measurement does not distinguish $|01\rangle$ from $|10\rangle$, so it does not collapse to one of them.

(C) This vector was not re-normalised; its length is $\sqrt{4/5} \neq 1$.

(D) $\frac{2}{5}$ uses the amplitudes instead of squares or counts only one of the two terms.

Remember. A coarse measurement keeps superpositions inside the measured subspace.

Q.30 Single-correct MCQ · Unitary vs. projector

Let $|u\rangle$ be a unit vector in a space of dimension $N \geq 2$. Which of these operators is *not* unitary?

(A) $H^{\otimes n}$ (for $N = 2^n$)
(B) ✓ $|u\rangle\langle u|$
(C) $I - 2\,|u\rangle\langle u|$
(D) $2\,|u\rangle\langle u| - I$

Answer: (B) **Detailed solution**

What you need to know. Unitary means $U^\dagger U = I$, and hence invertible. Projectors have eigenvalue 0.

Step-by-step solution. $|u\rangle\langle u|$ is a projector: it has eigenvalue 0, so it is not invertible and cannot be unitary. A reflection $R = \pm(I - 2\,|u\rangle\langle u|)$ is Hermitian with $R^2 = I$, so $R^\dagger R = I$. $H^{\otimes n}$ is a tensor product of unitaries.

Why the other options are wrong.

(A) A tensor product of unitaries is unitary.

(C) This is a reflection: Hermitian and squares to I, so it is unitary.

(D) This is also a reflection, the negative of the previous one.

Remember. Projectors are not gates; reflections are.

Q.31 Single-correct MCQ · Overlap $\cos\theta$

For $N = 16$ let $|s\rangle$ be the uniform superposition of all 16 basis states and $|r\rangle$ the uniform superposition of 15 of them (all except one). The value of $\langle r|s\rangle$ is

(A) ✓ $\dfrac{\sqrt{15}}{4}$ **(B)** $\dfrac{15}{16}$ **(C)** $\dfrac{1}{4}$ **(D)** $\dfrac{1}{\sqrt{15}}$

Answer: (A) **Detailed solution**

What you need to know. Overlap is the sum of products of matching amplitudes. Here $\cos\theta = \langle r|s\rangle$ and $\sin\theta = \langle m|s\rangle$.

Step-by-step solution. $\langle r|s\rangle = \sum_x \frac{1}{\sqrt{15}} \cdot \frac{1}{\sqrt{16}}$ over the 15 common basis states, which is $\frac{15}{\sqrt{15}\cdot 4} = \frac{\sqrt{15}}{4}$. It equals $\cos\theta$ with $\sin\theta = \frac{1}{4}$. $\frac{15}{16}$ is $\cos^2\theta$: the probability, not the overlap.

Why the other options are wrong.

(B) $\frac{15}{16}$ is $|\langle r|s\rangle|^2$, the probability.

(C) $\frac{1}{4} = \sin\theta$ is the overlap of $|s\rangle$ with the *marked* state.

(D) $\frac{1}{\sqrt{15}}$ is a single amplitude of $|r\rangle$.

Remember. The overlaps of $|s\rangle$ with the two axes define θ.

Q.32 Single-correct MCQ · State-vector memory

A state vector of $n = 33$ qubits is stored in double precision (16 bytes per complex amplitude). The memory needed is

(A) 256 GiB **(B)** 16 GiB **(C)** 64 GiB **(D)** ✓ 128 GiB

Answer: (D) Detailed solution

What you need to know. Memory $= 2^n \times$ bytes per amplitude, and 2^{30} bytes $= 1$ GiB.
Step-by-step solution. $2^{33} \times 16 = 2^{33} \times 2^4 = 2^{37}$ bytes $= 2^7 \times 2^{30}$ bytes $= 128$ GiB. Every extra qubit doubles it. (16 GiB is the value for $n = 30$.)
Why the other options are wrong.
(A) 256 GiB is the value for $n = 34$.
(B) 16 GiB is the value for $n = 30$.
(C) 64 GiB is the value for $n = 32$, or for $n = 33$ in single precision.
Remember. Memory doubles with every qubit: $n = 30$ needs 16 GiB in double precision.

Q.33 Single-correct MCQ · Diffusion for one qubit

For a single qubit ($n = 1$), the operator $2\,|+\rangle\langle+| - I$ equals

(A) ✓ X **(B)** H **(C)** Z **(D)** $-I$

Answer: (A) Detailed solution

What you need to know. $|+\rangle\langle+|$ has all entries $\frac{1}{2}$; the reflection about $|+\rangle$ fixes $|+\rangle$ and negates $|-\rangle$.
Step-by-step solution. $|+\rangle\langle+| = \frac{1}{2}\left(\begin{smallmatrix}1&1\\1&1\end{smallmatrix}\right)$, so $2\,|+\rangle\langle+| - I = \left(\begin{smallmatrix}1&1\\1&1\end{smallmatrix}\right) - \left(\begin{smallmatrix}1&0\\0&1\end{smallmatrix}\right) = \left(\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right) = X$. *Check:* $X\,|+\rangle = |+\rangle$ and $X\,|-\rangle = -\,|-\rangle$, which are the eigenvalues required of this reflection.
Why the other options are wrong.
(B) H does not fix $|+\rangle$: it maps $|+\rangle$ to $|0\rangle$.
(C) Z fixes $|0\rangle$ and negates $|1\rangle$: that is the reflection about $|0\rangle$.
(D) $-I$ would negate $|+\rangle$ too, not fix it.
Remember. Diffusion on one qubit is simply X.

Q.34 Single-correct MCQ · $|s\rangle$ is fixed by D

For n qubits let $|s\rangle = H^{\otimes n}\,|0\ldots0\rangle$ and $D = 2\,|s\rangle\langle s| - I$. If the oracle marks nothing (it acts as the identity), then after k iterations the state is

(A) $(-1)^k\,|s\rangle$ **(B)** $|0\ldots0\rangle$ for odd k
(C) a state whose amplitudes depend on k **(D)** ✓ $|s\rangle$ for every k

Answer: (D) Detailed solution

What you need to know. $|s\rangle$ is the $+1$ eigenvector of D.
Step-by-step solution. $D\,|s\rangle = 2\,|s\rangle\,\langle s|s\rangle - |s\rangle = |s\rangle$, so $|s\rangle$ is an eigenvector of D with eigenvalue $+1$. With $O = I$, each iteration is just D and leaves $|s\rangle$ unchanged. Measuring gives a uniformly random string, which matches $\sin^2\theta = 0$: nothing can be amplified when no item is marked.
Why the other options are wrong.
(A) The eigenvalue is $+1$, so no factor $(-1)^k$ arises. (The circuit form $-D$ would give such a factor, but it is a global phase.)
(B) D does not map $|s\rangle$ to $|0\ldots0\rangle$; that is what $H^{\otimes n}$ does.

(C) The amplitudes stay at $1/\sqrt{N}$ because $|s\rangle$ is an eigenvector.
Remember. Without a marked item the algorithm does nothing, as $\theta = 0$ predicts.

Q.35 MATCHING LISTS · Single-qubit gate actions

Match each expression in List-I with its value in List-II.

List-I		List-II	
(P)	$H\,\|1\rangle$	**(1)**	$-\,\|-\rangle$
(Q)	$Z\,\|-\rangle$	**(2)**	$\|0\rangle$
(R)	$X\,\|-\rangle$	**(3)**	$\|+\rangle$
(S)	$H\,\|+\rangle$	**(4)**	$\|-\rangle$

(A) P→4, Q→1, R→3, S→2
(B) P→4, Q→3, R→2, S→1
(C) ✓ P→4, Q→3, R→1, S→2
(D) P→3, Q→4, R→1, S→2

Answer: (C) **Detailed solution**

What you need to know. $H\,|0\rangle = |+\rangle$, $H\,|1\rangle = |-\rangle$, Z exchanges $|\pm\rangle$, and $X\,|\pm\rangle = \pm\,|\pm\rangle$.
Step-by-step solution. $H\,|1\rangle = |-\rangle$ (4). $Z\,|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = |+\rangle$ (3). $X\,|-\rangle = \frac{1}{\sqrt{2}}(|1\rangle - |0\rangle) = -\,|-\rangle$ (1). $H\,|+\rangle = |0\rangle$ because $H^2 = I$ (2). Hence P→4, Q→3, R→1, S→2.
Why the other options are wrong.
(A) This gives $Z\,|-\rangle = -\,|-\rangle$, which is the action of X; Z exchanges $|\pm\rangle$.
(B) This exchanges R and S: $X\,|-\rangle$ cannot be $|0\rangle$, and $H\,|+\rangle$ is $|0\rangle$.
(D) This exchanges P and Q: $H\,|1\rangle$ is $|-\rangle$, and $Z\,|-\rangle$ is $|+\rangle$.
Remember. Keep the table of H, X, Z on $|0\rangle, |1\rangle, |\pm\rangle$ at your fingertips.

Q.36 MATCHING LISTS · Product vs. entangled states

Match each two-qubit state in List-I with its description in List-II.

List-I		List-II	
(P)	$\frac{1}{2}(\|00\rangle + \|01\rangle + \|10\rangle + \|11\rangle)$	**(1)**	entangled, with four amplitudes of equal size
(Q)	$\frac{1}{2}(\|00\rangle - \|01\rangle - \|10\rangle + \|11\rangle)$	**(2)**	the product $\|-\rangle\,\|-\rangle$
(R)	$\frac{1}{2}(\|00\rangle + \|01\rangle + \|10\rangle - \|11\rangle)$	**(3)**	entangled, with only two non-zero amplitudes
(S)	$\frac{1}{\sqrt{2}}(\|00\rangle + \|11\rangle)$	**(4)**	the product $\|+\rangle\,\|+\rangle$

(A) P→4, Q→2, R→3, S→1
(B) ✓ P→4, Q→2, R→1, S→3
(C) P→4, Q→1, R→2, S→3
(D) P→2, Q→4, R→1, S→3

Answer: (B) **Detailed solution**

What you need to know. A two-qubit state is a product iff $c_{00}c_{11} = c_{01}c_{10}$.
Step-by-step solution. Test $c_{00}c_{11}$ against $c_{01}c_{10}$; they are equal exactly for a product state. P: $\frac{1}{4} = \frac{1}{4}$, product, equal to $|+\rangle\,|+\rangle$ (4). Q: $\frac{1}{4} = \frac{1}{4}$, product $|-\rangle\,|-\rangle$ (2). R: $-\frac{1}{4} \neq \frac{1}{4}$, entangled with four equal magnitudes (1); this is the state after the oracle for $n = 2$. S: $\frac{1}{2} \neq 0$, entangled with two non-zero amplitudes (3).
Why the other options are wrong.
(A) This exchanges R and S: state R has four non-zero amplitudes, and state S has two.

(C) This calls Q entangled, but it passes the product test and equals $|-\rangle\,|-\rangle$.

(D) This exchanges P and Q: all plus signs give $|+\rangle\,|+\rangle$.

Remember. One sign flip on $|s\rangle$ with $n = 2$ gives an entangled state, while Hadamards alone never do.

Q.37 Assertion–Reason · Why the oracle alone reveals nothing

Assertion (A): Measuring in the computational basis immediately after a phase oracle acts on $|s\rangle$ gives a uniformly random string; the oracle step alone reveals nothing.

Reason (R): The oracle multiplies each amplitude by ± 1, and the Born probabilities depend only on the magnitudes of the amplitudes.

(A) ✓ Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A

(B) Both A and R are true, but R is *not* the correct explanation of A

(C) A is true, but R is false

(D) A is false, but R is true

Answer: (A) Detailed solution

What you need to know. Probabilities depend only on $|c_x|^2$, while the oracle changes only signs.
Step-by-step solution. **R is true:** $O\,|x\rangle = (-1)^{f(x)}\,|x\rangle$ and $|\pm a|^2 = |a|^2$. **A is true** because of R: the magnitudes stay $1/\sqrt{N}$, so the distribution stays uniform. R is the correct explanation. Answer: (A).
Why the other options are wrong.

(B) R does explain A: it gives the exact reason that the distribution stays uniform.

(C) R is true: $|\pm a|^2 = |a|^2$.

(D) A is true: the output is uniformly random whichever item is marked.

Remember. The oracle's effect is invisible until an interfering step such as diffusion follows.

Q.38 Statement I / II · Relative phases are observable

Statement I: The states $|\psi\rangle$ and $-\,|\psi\rangle$ give identical statistics for every measurement.

Statement II: Multiplying only *one* basis-state amplitude of a superposition by -1 can never change the statistics of any measurement.

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) ✓ Statement I is true, but Statement II is false

(D) Statement I is false, but Statement II is true

Answer: (C) Detailed solution

What you need to know. Global phases are unobservable; relative phases can be observed after interference.
Step-by-step solution. **I is true:** a global phase cancels in $|\langle\phi|\psi\rangle|^2$ for every $|\phi\rangle$. **II is false:** that is a *relative* phase. Counterexample: $|+\rangle$ and $|-\rangle$ differ by the sign of $|1\rangle$ only, yet a measurement in the $\{|+\rangle, |-\rangle\}$ basis separates them perfectly. Grover's algorithm exists because the sign is detectable after interference. Answer: (C).
Why the other options are wrong.

(A) II is false: $|+\rangle$ and $|-\rangle$ differ only by the sign of one amplitude and yet are distinguishable.

(B) I is true: a global phase cancels in every probability.

(D) I is true, so this option fails.

Remember. A sign on one amplitude is a relative phase: it is exactly the signal that Grover amplifies.

Paragraph for Questions 39 and 40

Three qubits start in $|0\rangle\,|0\rangle\,|1\rangle$ (input qubits 1 and 2, ancilla qubit 3). A Hadamard gate is applied to all three. Then the Toffoli gate is applied with qubits 1 and 2 as controls and qubit 3 as target: it implements $U_f\,|x\rangle\,|y\rangle = |x\rangle\,|y \oplus f(x)\rangle$ with $f(x_1x_2) = x_1 \wedge x_2$.

Q.39 PARAGRAPH BASED · Toffoli as a phase oracle

The state of the three qubits after the Toffoli gate is

(A) $\frac{1}{2}(\,|00\rangle - |01\rangle - |10\rangle - |11\rangle\,)\,|-\rangle$

(B) $\frac{1}{2}(\,|00\rangle + |01\rangle + |10\rangle - |11\rangle\,)\,|+\rangle$

(C) $\frac{1}{2}(\,|00\rangle + |01\rangle + |10\rangle + |11\rangle\,)\,|-\rangle$

(D) ✓ $\frac{1}{2}(\,|00\rangle + |01\rangle + |10\rangle - |11\rangle\,)\,|-\rangle$

Answer: (D) Detailed solution

What you need to know. $H\,|1\rangle = |-\rangle$; kickback puts the sign $(-1)^{f(x)}$ on $|x\rangle$.

Step-by-step solution. After the Hadamards: $\frac{1}{2}\sum_x |x\rangle\,|-\rangle$, since $H\,|1\rangle = |-\rangle$. By kickback each term gets $(-1)^{f(x)}$, and $f = 1$ only for $x = 11$. So the state is $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle)\,|-\rangle$. The ancilla stays in $|-\rangle$, unentangled with the inputs.

Why the other options are wrong.

(A) Only $x = 11$ has $f = 1$, so only one sign flips.

(B) The ancilla remains in $|-\rangle$; it does not change to $|+\rangle$.

(C) This forgets the kicked-back sign on $|11\rangle$.

Remember. Toffoli plus ancilla $|-\rangle$ is the phase oracle for $x_1 \wedge x_2$.

Q.40 PARAGRAPH BASED · Ancilla stays unentangled

Now the ancilla alone is measured in the computational basis. The probability of the outcome 1, and the state of the input qubits afterwards, are

(A) $\frac{1}{4}$; the input qubits collapse to $|11\rangle$

(B) ✓ $\frac{1}{2}$; the input qubits remain in $\frac{1}{2}(\,|00\rangle + |01\rangle + |10\rangle - |11\rangle\,)$

(C) $\frac{1}{2}$; the input qubits collapse to $|11\rangle$

(D) 1; the input qubits remain in $\frac{1}{2}(\,|00\rangle + |01\rangle + |10\rangle - |11\rangle\,)$

Answer: (B) Detailed solution

What you need to know. A measurement on one factor of a product state does not disturb the other factor.

Step-by-step solution. The state after the gate is a product of the input state and $|-\rangle$ (see Q.39). Measuring $|-\rangle$ gives 1 with probability $\frac{1}{2}$, and because the ancilla is not entangled with the inputs, the input qubits are not disturbed.

Why the other options are wrong.

(A) This treats the ancilla as if it were in $|f(x)\rangle$, i.e. started in $|0\rangle$.

(C) The probability is right, but the inputs do not collapse because there is no entanglement.

(D) $|-\rangle$ gives outcome 1 with probability $\frac{1}{2}$, not 1.

Remember. With an ancilla in $|-\rangle$ the ancilla stays unentangled and can be re-used.

SECTION C · Grover Basics Q.41–60

Q.41 Single-correct MCQ · $N = 4$ is exact in one step

Grover's algorithm is run on $n = 2$ qubits ($N = 4$) with one marked item. After exactly one iteration, the probability of measuring the marked item is

(A) $\frac{1}{2}$ **(B)** $\frac{1}{4}$ **(C)** $\frac{3}{4}$ **(D)** ✓ 1

Answer: (D) Detailed solution

What you need to know. $\sin\theta = 1/\sqrt{N}$ and $P_k = \sin^2((2k+1)\theta)$. For $N = 4$, $\theta = 30°$.
Step-by-step solution. $\sin\theta = \frac{1}{\sqrt{4}} = \frac{1}{2}$, so $\theta = 30°$. After $k = 1$ iteration $P = \sin^2(3\theta) = \sin^2 90° = 1$. In amplitudes: after the oracle the entries are $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{2})$ with mean $\frac{1}{4}$; inversion about the mean gives $(0, 0, 0, 1)$.
Why the other options are wrong.
(A) $\frac{1}{2}$ is the result for $M = N/2$, not for this case.
(B) $\frac{1}{4}$ is the probability before any iteration.
(C) $\frac{3}{4}$ is the total probability of the unmarked items at the start.
Remember. $N = 4$ is the unique single-item case where one iteration is exact.

Q.42 Single-correct MCQ · One iteration, $N = 8$

For $N = 8$ with one marked item, the probability of measuring the marked item after one iteration is

(A) 1 **(B)** ✓ $\frac{25}{32}$ **(C)** $\frac{5}{8}$ **(D)** $\frac{9}{16}$

Answer: (B) Detailed solution

What you need to know. Trace the amplitudes with inversion about the mean, or use $P_1 = s(3 - 4s)^2$ with $s = 1/N$.
Step-by-step solution. Write $a = 1/\sqrt{8}$. After the oracle: seven entries a, one entry $-a$. Mean $= \frac{6a}{8} = \frac{3}{4}a$. The marked entry becomes $2 \cdot \frac{3}{4}a + a = \frac{5}{2}a$, so $P = \frac{25}{4}a^2 = \frac{25}{4} \cdot \frac{1}{8} = \frac{25}{32}$. *Check:* $\sin^2 3\theta = s(3 - 4s)^2$ with $s = \frac{1}{8}$ gives $\frac{1}{8} \cdot \frac{25}{4} = \frac{25}{32}$.
Why the other options are wrong.
(A) $P = 1$ happens for $N = 4$ only; here $3\theta \neq 90°$.
(C) $\frac{5}{8}$ uses the amplitude factor $\frac{5}{2}$ divided by 4 and forgets to square.
(D) $\frac{9}{16}$ uses the mean $\frac{3}{4}$ squared instead of the new amplitude.
Remember. For small N, inversion about the mean is the fastest method.

Q.43 SINGLE-CORRECT MCQ · k_{opt} and P for $n = 4$

For $n = 4$ qubits with one marked item ($\theta \approx 0.2527$ rad), the optimal number of iterations k_{opt} and the resulting success probability are

(A) $k_{\text{opt}} = 2$ and $P \approx 0.908$ **(B)** $k_{\text{opt}} = 4$ and $P \approx 0.582$

(C) $k_{\text{opt}} = 3$ and $P = 1$ **(D)** ✓ $k_{\text{opt}} = 3$ and $P \approx 0.961$

Answer: (D) Detailed solution

What you need to know. $k_{\text{opt}} = \text{round}\left(\frac{\pi}{4\theta} - \frac{1}{2}\right)$ and $P = \sin^2((2k+1)\theta)$.
Step-by-step solution. $\frac{\pi}{4\theta} - \frac{1}{2} = \frac{0.7854}{0.2527} - 0.5 = 3.108 - 0.5 = 2.608$, which rounds to $k_{\text{opt}} = 3$. Then $P = \sin^2(7\theta) = \sin^2(1.769) \approx 0.961$. The angle 7θ does not hit $\frac{\pi}{2}$ exactly, so $P < 1$. For comparison, $k = 2$ gives $\sin^2 5\theta \approx 0.908$ and $k = 4$ gives $\sin^2 9\theta \approx 0.582$.
Why the other options are wrong.
(A) $k = 2$ is below the optimum: 5θ is further from $\frac{\pi}{2}$ than 7θ.
(B) $k = 4$ over-rotates, and P falls to 0.58.
(C) $k = 3$ is right but $P < 1$: $7\theta = 1.769$ rad, not $\frac{\pi}{2}$.
Remember. Optimal does not mean perfect: P is close to 1 but usually not equal to 1.

Q.44 SINGLE-CORRECT MCQ · k_{opt} for $n = 15$

The accompanying notebook searches $2^{15} = 32\,768$ items for one marked item. The optimal number of iterations is

(A) 181 **(B)** ✓ 142 **(C)** 16 384 **(D)** 128

Answer: (B) Detailed solution

What you need to know. $k_{\text{opt}} \approx \frac{\pi}{4}\sqrt{N}$. Note $\sqrt{32768} \approx 181$.
Step-by-step solution. $\sqrt{32768} = 181.02$, $\theta = \arcsin(1/181.02) = 0.005524$, so $\frac{\pi}{4\theta} - \frac{1}{2} = 142.17 - 0.5 = 141.67$, which rounds to 142. 181 forgets the factor $\frac{\pi}{4}$, and $16\,384 = N/2$ is the classical average.
Why the other options are wrong.
(A) $181 = \sqrt{N}$ forgets the factor $\frac{\pi}{4}$.
(C) $16\,384 = N/2$ is the classical average.
(D) 128 is an unrelated power of 2.
Remember. The rule of thumb is $0.785\sqrt{N}$.

Q.45 SINGLE-CORRECT MCQ · Building the oracle

An oracle for the marked string $|0101\rangle$ is built as: X gates, then the multi-controlled Z on all four qubits, then the same X gates. The X gates act on

(A) qubits 2 and 4 **(B)** no qubit **(C)** ✓ qubits 1 and 3 **(D)** all four qubits

Answer: (C) Detailed solution

What you need to know. The multi-controlled Z flips $|1\ldots1\rangle$ only, so the X gates must map the marked string to all ones.
Step-by-step solution. The multi-controlled Z flips the sign of $|1111\rangle$ only, so the X layer must map the marked string to $|1111\rangle$. That means flipping exactly the bits where x^* has a 0. In 0101 these are qubits 1 and 3. The second X layer undoes the first, so the sign flip ends up on $|0101\rangle$.
Why the other options are wrong.
(A) Qubits 2 and 4 hold 1 already: flipping them maps $|1010\rangle$ to all ones, the wrong item.

(B) With no X gates the oracle marks $|1111\rangle$.

(D) Flipping all qubits marks the complement $|1010\rangle$.

Remember. Rule: X on the zeros of x^*.

Q.46 SINGLE-CORRECT MCQ · Gate count per iteration

Use the circuit in the conventions box and count each multi-controlled Z as one gate. The number of gates in *one* Grover iteration for $n = 6$, $x^* = 010011$ is

(A) ✓ 32 **(B)** 30 **(C)** 26 **(D)** 38

Answer: (A) Detailed solution

What you need to know. Per iteration: oracle $2z+1$ gates and diffusion $4n+1$ gates, so $4n+2z+2$ in total.

Step-by-step solution. The oracle has $2z$ X gates, where z is the number of 0s in x^* (here $z = 3$), and one multi-controlled Z: $6+1 = 7$ gates. The diffusion has $2n$ H, $2n$ X and one multi-controlled Z: $12 + 12 + 1 = 25$ gates. Total $= 4n + 2z + 2 = 24 + 6 + 2 = 32$.

Why the other options are wrong.

(B) 30 omits the two multi-controlled Z gates.

(C) 26 omits the $2z$ oracle X gates.

(D) 38 counts $2n$ oracle X gates, as if every bit were 0.

Remember. The formula $4n + 2z + 2$ is the one the notebook uses in its time estimate.

Q.47 SINGLE-CORRECT MCQ · Diffusion circuit $= -D$

Let C be the multi-controlled Z on n qubits. The circuit $H^{\otimes n}X^{\otimes n}\,C\,X^{\otimes n}H^{\otimes n}$ implements

(A) $2\,|s\rangle\langle s| - I = D$ **(B)** ✓ $I - 2\,|s\rangle\langle s| = -D$

(C) $I - 2\,|0\ldots0\rangle\langle 0\ldots0|$ **(D)** $I - 2\,|1\ldots1\rangle\langle 1\ldots1|$

Answer: (B) Detailed solution

What you need to know. $U(I - 2\,|a\rangle\langle a|)U^\dagger = I - 2U\,|a\rangle\langle a|\,U^\dagger$; a reflection conjugated by a unitary is a reflection about the image.

Step-by-step solution. $C = I - 2\,|1\ldots1\rangle\langle 1\ldots1|$. Conjugating by $X^{\otimes n}$ gives $I - 2\,|0\ldots0\rangle\langle 0\ldots0|$, and conjugating by $H^{\otimes n}$ gives $I - 2\,|s\rangle\langle s|$, since $H^{\otimes n}\,|0\ldots0\rangle = |s\rangle$. That is $-D$: the circuit equals the diffusion up to the global phase -1, which has no observable effect.

Why the other options are wrong.

(A) This is the intended operator, but the circuit gives its negative. Only a global phase separates them, but the exact identity is $-D$.

(C) This is the operator at an intermediate step, before the H layers are included.

(D) This is the multi-controlled Z itself.

Remember. Diffusion circuit $= -D$, and the minus sign is a harmless global phase.

Q.48 Single-correct MCQ · Unmarked amplitude

For $N = 16$ with one marked item, the amplitude of each *unmarked* item after one iteration is

(A) $\frac{1}{4}$ **(B)** $\frac{11}{16}$ **(C)** ✓ $\frac{3}{16}$ **(D)** $\frac{7}{32}$

Answer: (C) **Detailed solution**

What you need to know. Inversion about the mean: new amplitude $= 2 \times$ mean $-$ old, with the mean taken after the oracle.

Step-by-step solution. Start with amplitudes $\frac{1}{4}$. After the oracle the mean is $\frac{15-1}{16} \cdot \frac{1}{4} = \frac{7}{32}$. An unmarked entry becomes $2 \cdot \frac{7}{32} - \frac{1}{4} = \frac{7}{16} - \frac{4}{16} = \frac{3}{16}$, and the marked entry becomes $\frac{7}{16} + \frac{4}{16} = \frac{11}{16}$. *Check:* $15 \cdot \frac{9}{256} + \frac{121}{256} = 1$.

Why the other options are wrong.

(A) $\frac{1}{4}$ is the initial amplitude.

(B) $\frac{11}{16}$ is the amplitude of the marked item.

(D) $\frac{7}{32}$ is the mean, not the new amplitude.

Remember. Unmarked amplitudes shrink a little and the marked one grows a lot.

Q.49 Single-correct MCQ · $M = N/4$

Let $N = 16$ with $M = 4$ marked items. After one iteration the probability of measuring a marked item is

(A) $\frac{3}{4}$ **(B)** $\frac{25}{32}$ **(C)** ✓ 1 **(D)** $\frac{1}{4}$

Answer: (C) **Detailed solution**

What you need to know. Several solutions: $\sin^2\theta = M/N$. $M = N/4$ gives $\theta = 30°$.

Step-by-step solution. $\sin^2\theta = \frac{M}{N} = \frac{1}{4}$, so $\theta = 30°$ and $P_1 = \sin^2 3\theta = \sin^2 90° = 1$. With $M = N/4$ a single iteration succeeds with certainty, for every N for which $M = N/4$ is an integer. The final state is the uniform superposition of the four marked items.

Why the other options are wrong.

(A) $\frac{3}{4}$ is the probability of the unmarked set before any iteration.

(B) $\frac{25}{32}$ is the value for $s = \frac{1}{8}$ and not for $s = \frac{1}{4}$.

(D) $\frac{1}{4}$ is the probability before any iteration.

Remember. $M = N/4$ is the special ratio for a one-step exact search.

Q.50 Single-correct MCQ · $M = N/2$

Let $N = 8$ with $M = 4$ marked items. The probability of a marked outcome after k iterations is

(A) 1 for even k and 0 for odd k **(B)** 0 for $k = 1$

(C) 1 for $k = 1$ **(D)** ✓ $\frac{1}{2}$ for every k

Answer: (D) **Detailed solution**

What you need to know. $M = N/2$ gives $\theta = 45°$, and $2\theta = 90°$ steps skip between the diagonals.

Step-by-step solution. $\sin^2\theta = \frac{1}{2}$ gives $\theta = 45°$, and $P_k = \sin^2((2k+1)45°)$. The angles $45°, 135°, 225°, \ldots$ all have $\sin^2 = \frac{1}{2}$. The iteration rotates by $2\theta = 90°$, which exchanges the two diagonals and never brings the state closer to the marked axis. Grover gives no gain at $M = N/2$.

Why the other options are wrong.

(A) The angles $45°, 135°, \ldots$ all have $\sin^2 = \frac{1}{2}$, so there is no oscillation between 0 and 1.

(B) That would need $3\theta = 180°$, i.e. $\theta = 60°$.

(C) That would need $3\theta = 90°$, i.e. $\theta = 30°$.

Remember. At $M = N/2$, classical guessing is already as good as Grover.

Q.51 Single-correct MCQ · $M = 3N/4$

Let $N = 8$ with $M = 6$ marked items. The probability of a marked outcome after one iteration is

(A) $\frac{3}{4}$ **(B)** $\frac{1}{4}$ **(C)** 1 **(D)** ✓ 0

Answer: (D) **Detailed solution**

What you need to know. $M > N/2$ gives $\theta > 45°$, and iterations can rotate past the marked axis at once.

Step-by-step solution. $\sin^2\theta = \frac{3}{4}$ gives $\theta = 60°$, so $P_1 = \sin^2 180° = 0$. Amplitudes confirm this: with $a = 1/\sqrt{8}$ the oracle gives six entries $-a$ and two entries $+a$; the mean is $-\frac{1}{2}a$; the marked entries become $-a + a = 0$. Measuring *without* any iteration ($P_0 = \frac{3}{4}$) is far better.

Why the other options are wrong.

(A) $\frac{3}{4}$ is P_0, the probability with no iteration.

(B) $\frac{1}{4}$ is the initial probability of the unmarked items.

(C) 1 would need $3\theta = 90°$; here $3\theta = 180°$.

Remember. If most items are marked, do not iterate: just measure.

Q.52 Single-correct MCQ · Over-rotation at $2k_{\text{opt}}$

On 2^{15} items with one marked item, $k_{\text{opt}} = 142$. If $k = 284$ iterations are used instead, the probability of the marked item is about

(A) ✓ 3×10^{-6} **(B)** $\frac{1}{2}$ **(C)** $\frac{1}{32768}$ **(D)** 1

Answer: (A) **Detailed solution**

What you need to know. P_k is periodic: at $k \approx 2k_{\text{opt}}$ the state has rotated by nearly π and returns to near the unmarked axis.

Step-by-step solution. The angle is $(2k+1)\theta = 569 \times 0.0055243 = 3.1433$ rad, which is 0.0017 past π. So $P = \sin^2(569\theta) \approx (0.0017)^2 \approx 3 \times 10^{-6}$. This is over-rotation: twice the optimal number of iterations brings the state back near the unmarked axis, and the marked item is then *less* likely than a random guess ($1/32768 \approx 3 \times 10^{-5}$).

Why the other options are wrong.

(B) $\frac{1}{2}$ is the value around $k \approx k_{\text{opt}}/2$, or $k \approx \frac{3}{2}k_{\text{opt}}$.

(C) $\frac{1}{32768} \approx 3 \times 10^{-5}$ is the starting value at $k = 0$, about ten times larger than the value at $k = 284$.

(D) 1 would be right near k_{opt}, not near $2k_{\text{opt}}$.

Remember. More iterations are not always better: Grover is a rotation, not a monotonic filter.

Q.53 Single-correct MCQ · Size of the speed-up

For $n = 20$ qubits with one marked item, Grover needs 804 iterations. A classical random search needs on average about 2^{19} queries. The classical cost divided by the quantum cost is about

(A) ✓ 650 **(B)** 2 **(C)** 1000 **(D)** 20

Answer: (A) Detailed solution

What you need to know. Ratio $= \frac{N/2}{(\pi/4)\sqrt{N}} = \frac{2}{\pi}\sqrt{N}$.

Step-by-step solution. $2^{19} = 524\,288$ and $524\,288/804 \approx 652$. In general the ratio is $\frac{N/2}{(\pi/4)\sqrt{N}} = \frac{2}{\pi}\sqrt{N}$, and here $\frac{2}{\pi} \times 1024 \approx 652$. The speed-up is quadratic, not exponential.

Why the other options are wrong.

(B) 2 would be a constant speed-up.

(C) 1000 rounds $\sqrt{N} = 1024$ and forgets the factor $\frac{2}{\pi}$.

(D) 20 is n; the ratio is not polynomial in n.

Remember. Quadratic speed-up: the gain is $\frac{2}{\pi}\sqrt{N}$.

Q.54 Single-correct MCQ · Shot statistics

A Grover circuit with success probability $P = 0.9613$ is run for 2000 shots. The expected marked count and its binomial standard deviation are about

(A) 1923 ± 44 **(B)** ✓ 1923 ± 9 **(C)** 2000 ± 0 **(D)** 1923 ± 77

Answer: (B) Detailed solution

What you need to know. Binomial: mean SP, standard deviation $\sqrt{SP(1-P)}$.

Step-by-step solution. Mean $= SP = 2000 \times 0.9613 \approx 1923$. Standard deviation $= \sqrt{SP(1-P)} = \sqrt{2000 \times 0.9613 \times 0.0387} = \sqrt{74.4} \approx 8.6$. $\pm 44 = \sqrt{1923}$ is the Poisson estimate, which is wrong here because P is close to 1.

Why the other options are wrong.

(A) $\pm 44 = \sqrt{1923}$ uses a Poisson estimate and ignores the factor $1 - P$.

(C) $P < 1$, so not every shot succeeds.

(D) $\pm 77 = S(1-P)$ is the expected number of failures, not a spread.

Remember. When P is close to 1, the spread is small: compare measurement against $\sqrt{SP(1-P)}$.

Q.55 Matching lists · k_{opt} against n

Match the number of qubits (one marked item) in List-I with k_{opt} in List-II. Use $k_{\text{opt}} = \text{round}\left(\frac{\pi}{4\theta} - \frac{1}{2}\right)$.

List-I		List-II	
(P)	$n = 2$	**(1)**	12
(Q)	$n = 6$	**(2)**	25
(R)	$n = 8$	**(3)**	1
(S)	$n = 10$	**(4)**	6

(A) ✓ P→3, Q→4, R→1, S→2 **(B)** P→3, Q→1, R→4, S→2

(C) P→3, Q→4, R→2, S→1 **(D)** P→4, Q→3, R→1, S→2

Answer: (A) Detailed solution

What you need to know. $k_{\text{opt}} \approx \frac{\pi}{4}2^{n/2}$, with exact rounding from $\frac{\pi}{4\theta} - \frac{1}{2}$.
Step-by-step solution. $n = 2$: $\theta = \frac{\pi}{6}$, $\frac{\pi}{4\theta} - \frac{1}{2} = 1$ (3). $n = 6$: $\theta \approx 0.1253$, giving $5.77 \to 6$ (4). $n = 8$: $\theta \approx 0.0625$, giving $12.06 \to 12$ (1). $n = 10$: $\theta \approx 0.03126$, giving $24.63 \to 25$ (2). Hence P→3, Q→4, R→1, S→2.
Why the other options are wrong.

(B) This exchanges Q and R: the iterations grow with n, so $n = 6$ cannot need more than $n = 8$.

(C) This exchanges R and S: $n = 8$ gives 12 and $n = 10$ gives 25.

(D) This exchanges P and Q: $n = 2$ gives $k = 1$ and $n = 6$ gives $k = 6$.

Remember. Adding 2 qubits roughly doubles k_{opt}: $6, 12, 25$.

Q.56 MATCHING LISTS · Blocks of the circuit

Match each part of the Grover circuit in List-I with its effect in List-II.

List-I		List-II	
(P)	$H^{\otimes n}$ on $\lvert 0\ldots0\rangle$	**(1)**	returns one bit string with its Born probability
(Q)	X gates on the 0 bits of x^*, multi-controlled Z, the same X gates	**(2)**	creates the uniform superposition $\lvert s\rangle$
(R)	$H^{\otimes n}X^{\otimes n}$, multi-controlled Z, $X^{\otimes n}H^{\otimes n}$	**(3)**	reflects about $\lvert s\rangle$, up to a global phase
(S)	measuring all qubits	**(4)**	flips the sign of $\lvert x^*\rangle$ only

(A) ✓ P→2, Q→4, R→3, S→1
(B) P→2, Q→4, R→1, S→3
(C) P→4, Q→2, R→3, S→1
(D) P→2, Q→3, R→4, S→1

Answer: (A) Detailed solution

What you need to know. The circuit has four distinct stages: preparation, oracle, diffusion and measurement.
Step-by-step solution. P: $H^{\otimes n}\lvert 0\ldots0\rangle = \lvert s\rangle$ (2). Q: the oracle recipe of Q.45 flips only $\lvert x^*\rangle$ (4). R: this is the diffusion circuit of Q.47 (3). S: the measurement samples one string from $\lvert c_x\rvert^2$ (1). Hence P→2, Q→4, R→3, S→1.
Why the other options are wrong.

(B) This exchanges R and S: the diffusion is unitary and measurement is not.

(C) This exchanges P and Q: the Hadamard layer prepares $\lvert s\rangle$, and the X-MCZ-X block marks.

(D) This exchanges Q and R: the block with H gates is the diffusion, not the oracle.

Remember. Oracle: reflect about the marked axis. Diffusion: reflect about $\lvert s\rangle$.

Q.57 Assertion–Reason · Independence of the marked item

Assertion (A): The success probability of Grover's algorithm does not depend on which single item is marked.

Reason (R): The oracle is a diagonal matrix in the computational basis.

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A

(B) ✓ Both A and R are true, but R is *not* the correct explanation of A

(C) A is true, but R is false

(D) A is false, but R is true

Answer: (B) Detailed solution

What you need to know. Decide A and R first, then ask whether R causes A. The cause here is the symmetry of $|s\rangle$.

Step-by-step solution. **A is true:** only $\langle x^*|s\rangle = 1/\sqrt{N}$ enters the analysis, and this is the same for every x^*. **R is true:** $O = \mathrm{diag}\left((-1)^{f(x)}\right)$. However, being diagonal does not explain A: an arbitrary diagonal gate need not treat all items alike. The explanation is the symmetry of $|s\rangle$, which has the same overlap with every basis state. Answer: (B).

Why the other options are wrong.

(A) R is true but is not the explanation: not every diagonal gate treats all items equally.

(C) R is true: O is diagonal with entries ± 1.

(D) A is true: $\langle x^*|s\rangle = 1/\sqrt{N}$ for every x^*.

Remember. True does not imply explanatory: in Assertion–Reason questions, check for a causal link.

Q.58 Statement I / II · Oscillation and $\sqrt{N}$ scaling

Statement I: In Grover's algorithm, applying more iterations never lowers the success probability.

Statement II: For one marked item, the optimal number of iterations grows as $\sqrt{N}$.

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true, but Statement II is false

(D) ✓ Statement I is false, but Statement II is true

Answer: (D) Detailed solution

What you need to know. P_k oscillates; $k_{\text{opt}} \propto \sqrt{N}$.

Step-by-step solution. **I is false:** $P_k = \sin^2((2k+1)\theta)$ oscillates. For $N = 4$ we have $P_1 = 1$ but $P_2 = \sin^2 150^\circ = \frac{1}{4}$. **II is true:** $k_{\text{opt}} \approx \frac{\pi}{4\theta} \approx \frac{\pi}{4}\sqrt{N}$. Answer: (D).

Why the other options are wrong.

(A) I is false: over-rotation is real, for instance for $N = 4$, $k = 2$.

(B) II is true: $k_{\text{opt}} \approx \frac{\pi}{4}\sqrt{N}$.

(C) I is false and II is true, which is the opposite of this option.

Remember. One counterexample is enough to reject a "never" statement.

Paragraph for Questions 59 and 60

Grover's algorithm runs on $n = 3$ qubits with the single marked item $x^* = |101\rangle$. It starts from the uniform state $|s\rangle$, in which every amplitude is $a = 1/\sqrt{8}$. One iteration is the oracle O followed by the diffusion D. Trace the amplitudes by hand with inversion about the mean.

Q.59 PARAGRAPH BASED · Amplitude trace, first iteration

After the first iteration the amplitudes are

(A) marked $\frac{3}{2}a$; each unmarked $\frac{1}{2}a$
(B) marked $\frac{7}{4}a$; each unmarked $\frac{3}{4}a$
(C) ✓ marked $\frac{5}{2}a$; each unmarked $\frac{1}{2}a$
(D) marked $\frac{5}{2}a$; each unmarked $\frac{3}{4}a$

Answer: (C) Detailed solution

What you need to know. Inversion about the mean with the mean computed after the oracle.
Step-by-step solution. After O: seven entries a and one entry $-a$; the mean is $\frac{7a-a}{8} = \frac{3}{4}a$. Inversion about the mean gives: unmarked $2 \cdot \frac{3}{4}a - a = \frac{1}{2}a$; marked $2 \cdot \frac{3}{4}a + a = \frac{5}{2}a$. *Check:* $(7 \cdot \frac{1}{4} + \frac{25}{4})a^2 = 8a^2 = 1$.
Why the other options are wrong.
(A) $\frac{3}{2}a = 2\times$mean omits the term $+a$ for the marked item: $2 \cdot \frac{3}{4}a - (-a) = \frac{5}{2}a$.
(B) $\frac{7}{4}a$ and $\frac{3}{4}a$ use the mean as the amplitude and do not preserve the length.
(D) The unmarked value is wrong: $\frac{3}{4}a$ is the mean itself.
Remember. Marked: $2\text{mean} + a$. Unmarked: $2\text{mean} - a$.

Q.60 PARAGRAPH BASED · Amplitude trace, second iteration

After the second iteration the probability of measuring $|101\rangle$ is

(A) $\frac{25}{32}$ **(B)** $\frac{11}{16}$ **(C)** ✓ $\frac{121}{128}$ **(D)** 1

Answer: (C) Detailed solution

What you need to know. Iterate the same two steps; the second mean is smaller because the marked amplitude is now large.
Step-by-step solution. Start from the result of Q.59: unmarked $\frac{1}{2}a$, marked $\frac{5}{2}a$. After O the marked entry is $-\frac{5}{2}a$, and the mean is $\frac{7 \cdot \frac{1}{2}a - \frac{5}{2}a}{8} = \frac{a}{8}$. The marked entry becomes $2 \cdot \frac{a}{8} + \frac{5}{2}a = \frac{11}{4}a$, so $P = \frac{121}{16}a^2 = \frac{121}{16 \cdot 8} = \frac{121}{128}$. (Each unmarked entry becomes $\frac{a}{4} - \frac{a}{2} = -\frac{a}{4}$.) *Check:* $P_2 = \sin^2 5\theta \approx 0.945$ with $\sin^2\theta = \frac{1}{8}$. A third iteration would over-rotate.
Why the other options are wrong.
(A) $\frac{25}{32}$ is the value after the *first* iteration.
(B) $\frac{11}{16}$ uses $\frac{11}{4}$ without squaring correctly with $a^2 = \frac{1}{8}$.
(D) $P = 1$ would need $5\theta = 90°$ exactly; here it is $103.5°$.
Remember. $k = 2$ is optimal for $N = 8$, with $P = \frac{121}{128} \approx 0.945$.

SECTION D · Advanced Grover Q.61–80

Q.61 Single-correct MCQ · Several solutions: k_{opt}

For $N = 1024$ with $M = 16$ marked items, the optimal number of iterations is

(A) 25 **(B)** ✓ 6 **(C)** 5 **(D)** 8

Answer: (B) Detailed solution

What you need to know. With M solutions $\sin\theta = \sqrt{M/N}$, and $k_{\text{opt}} \approx \frac{\pi}{4}\sqrt{N/M}$.
Step-by-step solution. $\sin\theta = \sqrt{16/1024} = \frac{1}{8}$, so $\theta = 0.12533$ rad. Then $\frac{\pi}{4\theta} - \frac{1}{2} = 6.267 - 0.5 = 5.767$, which rounds to 6. The success probability is $\sin^2 13\theta \approx 0.9966$. (25 is the value for one marked item; $8 = \sqrt{N/M}$ forgets $\frac{\pi}{4}$.)
Why the other options are wrong.
(A) 25 is the value for $M = 1$ and $N = 1024$.
(C) 5 rounds down instead of to the nearest integer: 5.77 rounds to 6.
(D) $8 = \sqrt{N/M}$ forgets $\frac{\pi}{4}$.
Remember. More solutions mean fewer iterations: divide N by M before taking the root.

Q.62 Single-correct MCQ · BBHT threshold for unknown M

When M is unknown, the algorithm of Boyer, Brassard, Høyer and Tapp picks k uniformly from $\{0, \ldots, m-1\}$. The average success probability is at least $\frac{1}{4}$ once $m \geq 1/\sin 2\theta$. For $N = 2^{20}$ and $M = 1$ this threshold on m is about

(A) ✓ 512 **(B)** 1024 **(C)** 804 **(D)** 2^{20}

Answer: (A) Detailed solution

What you need to know. The BBHT lemma: the average of P_k over $k < m$ is $\frac{1}{2} - \frac{\sin 4m\theta}{4m\sin 2\theta}$.
Step-by-step solution. $\sin\theta = 2^{-10}$ is small, so $\sin 2\theta \approx 2\sin\theta = 2^{-9}$ and $\frac{1}{\sin 2\theta} \approx 2^9 = 512$. In general the threshold is about $\frac{1}{2}\sqrt{N/M}$, which comes from the identity $\frac{1}{m}\sum_{k=0}^{m-1}\sin^2((2k+1)\theta) = \frac{1}{2} - \frac{\sin 4m\theta}{4m\sin 2\theta}$; the last term is at most $\frac{1}{4}$ in size once $m \geq 1/\sin 2\theta$.
Why the other options are wrong.
(B) $1024 = \sqrt{N}$ forgets $\sin 2\theta \approx 2\sin\theta$.
(C) $804 \approx \frac{\pi}{4}\sqrt{N}$ is k_{opt}, not the threshold.
(D) $2^{20} = N$ would destroy the speed-up.
Remember. Random k below about $\frac{1}{2}\sqrt{N/M}$ already gives average success $\geq \frac{1}{4}$.

Q.63 Single-correct MCQ · Exact Grover search

To obtain success probability exactly 1, the angle is reduced from θ to θ' so that $(2k+1)\theta' = \pi/2$ with $k = \lceil \frac{\pi}{4\theta} - \frac{1}{2} \rceil$. For $N = 16$, $M = 1$ this requires

(A) $k = 3$ and $\theta' = \pi/12$ **(B)** $k = 4$ and $\theta' = \pi/18$
(C) $k = 2$ and $\theta' = \pi/10$ **(D)** ✓ $k = 3$ and $\theta' = \pi/14$

Answer: (D) Detailed solution

What you need to know. An exact search needs $(2k+1)\theta' = \frac{\pi}{2}$ with $\theta' \leq \theta$, so round k *up*.
Step-by-step solution. $\frac{\pi}{4\theta} - \frac{1}{2} = 2.608$, so the ceiling is $k = 3$. Then $7\theta' = \frac{\pi}{2}$ gives $\theta' = \frac{\pi}{14} \approx 0.2244$, slightly below $\theta \approx 0.2527$ as required. One way to reduce the angle is to mark only with amplitude $\sin\theta'/\sin\theta$ by adding an ancilla rotation. The ceiling matters: rounding down to $k = 2$ would need

$\theta' = \frac{\pi}{10} > \theta$, which cannot be reached by reducing the marked weight.

Why the other options are wrong.

(A) $\theta' = \frac{\pi}{12}$ gives $7\theta' = \frac{7\pi}{12} \neq \frac{\pi}{2}$.

(B) $k = 4$ also satisfies the equation with $\theta' = \frac{\pi}{18}$, but it is not the ceiling and costs one more iteration.

(C) $k = 2$ needs $\theta' = \frac{\pi}{10} > \theta$, and the angle cannot be increased by reducing the marked weight.

Remember. Round up, then shrink the angle a little, and the probability becomes exactly 1.

Q.64 SINGLE-CORRECT MCQ · Optimality (BBBV, Zalka)

Which statement about the query complexity of unstructured search is correct?

(A) Grover's algorithm is optimal only when N is a power of 2

(B) Quantum algorithms need only $O(\log N)$ queries, because $n = \log_2 N$ qubits hold all N items

(C) The lower bound $\Omega(\sqrt{N})$ holds only for classical algorithms

(D) ✓ Every quantum algorithm needs $\Omega(\sqrt{N})$ oracle queries, so Grover's algorithm is optimal up to a constant

Answer: (D) Detailed solution

What you need to know. BBBV lower bound: $\Omega(\sqrt{N})$ queries for black-box search. Zalka: $\frac{\pi}{4}\sqrt{N}$ is exactly optimal.

Step-by-step solution. The lower bound of Bennett, Bernstein, Brassard and Vazirani states that any quantum algorithm with black-box access needs $\Omega(\sqrt{N})$ queries; Zalka showed that $\frac{\pi}{4}\sqrt{N}$ is optimal even including the constant. Superposition of N items does not give $O(\log N)$ search, since a measurement returns only one string and the marked amplitude grows slowly.

Why the other options are wrong.

(A) The result does not depend on N being a power of 2.

(B) Holding N amplitudes in n qubits does not allow reading them out, and $\log N$ queries would violate the lower bound.

(C) Classical search needs $\Theta(N)$; the bound $\Omega(\sqrt{N})$ is the quantum one.

Remember. No black-box quantum search can beat $\sqrt{N}$ by more than a constant.

Q.65 SINGLE-CORRECT MCQ · Parallel search

The search space of size N is split evenly among 16 independent quantum computers, each running Grover on its share. Compared with one computer on the whole space, the wall-clock time is divided by

(A) 16, and the total number of oracle queries is unchanged

(B) 16, and the total number of oracle queries is multiplied by 4

(C) ✓ 4, and the total number of oracle queries is multiplied by 4

(D) 4, and the total number of oracle queries is unchanged

Answer: (C) Detailed solution

What you need to know. Time on each machine $\propto \sqrt{N/p}$; total queries $\propto p\sqrt{N/p} = \sqrt{Np}$.

Step-by-step solution. Each computer searches $N/16$ items: time $\frac{\pi}{4}\sqrt{N/16} = \frac{1}{4} \cdot \frac{\pi}{4}\sqrt{N}$, so the time falls by $\sqrt{16} = 4$. Total queries: $16 \times \frac{1}{4} = 4$ times the single-computer count. Grover parallelises only with a factor $\sqrt{p}$, in contrast to the factor p for classical search.

Why the other options are wrong.

(A) The factor 16 would be the classical speed-up.

(B) Time falls by $\sqrt{16} = 4$, not by 16.

(D) The time factor is right, but the total increases: $16 \times \frac{1}{4} = 4$.

Remember. Grover parallelises poorly: p machines give only $\sqrt{p}$.

Q.66 SINGLE-CORRECT MCQ · Amplitude amplification cost

Amplitude amplification boosts a procedure $\mathcal{A}$ that succeeds with probability $a = 0.01$ ($\theta \approx 0.10017$). At the optimal k, one uses $\mathcal{A}$ or $\mathcal{A}^{-1}$ in total

(A) 7 times **(B)** ✓ 15 times **(C)** 100 times **(D)** 8 times

Answer: (B) Detailed solution

What you need to know. $Q = -\mathcal{A}S_0\mathcal{A}^{-1}S_\chi$: each iteration uses $\mathcal{A}$ and $\mathcal{A}^{-1}$ once, plus one initial $\mathcal{A}$.

Step-by-step solution. Optimal k: $\frac{\pi}{4\theta} - \frac{1}{2} = 7.34$, so $k = 7$. The initial state uses $\mathcal{A}$ once, and each iteration $Q = -\mathcal{A}S_0\mathcal{A}^{-1}S_\chi$ uses $\mathcal{A}$ and $\mathcal{A}^{-1}$ once each. Total $2k + 1 = 15$ uses. Classical repetition needs about $1/a = 100$ runs on average.

Why the other options are wrong.

(A) 7 is the iteration count, not the number of uses of $\mathcal{A}$.

(C) 100 is the classical repetition count.

(D) 8 is $k + 1$ and ignores the two uses per iteration.

Remember. $2k + 1$ uses of $\mathcal{A}$, compared with $1/a$ classical repetitions.

Q.67 SINGLE-CORRECT MCQ · Eigenphases of G

For $N = 16$, $M = 4$, the phases $\varphi \in [0, 1)$ of the eigenvalues $e^{2\pi i\varphi}$ of G on the plane containing $|s\rangle$ are

(A) $\frac{1}{4}$ and $\frac{3}{4}$ **(B)** $\frac{1}{3}$ and $\frac{2}{3}$ **(C)** $\frac{1}{12}$ and $\frac{11}{12}$ **(D)** ✓ $\frac{1}{6}$ and $\frac{5}{6}$

Answer: (D) Detailed solution

What you need to know. G rotates by 2θ: eigenvalues $e^{\pm 2i\theta}$, so $\varphi = \theta/\pi$ and $1 - \theta/\pi$.

Step-by-step solution. $\sin^2\theta = \frac{1}{4}$ gives $\theta = \frac{\pi}{6}$. G rotates by 2θ, so its eigenvalues are $e^{\pm 2i\theta} = e^{\pm i\pi/3}$. Writing $e^{2\pi i\varphi}$: $\varphi = \frac{\theta}{\pi} = \frac{1}{6}$, and $1 - \frac{1}{6} = \frac{5}{6}$. $\frac{1}{12}$ confuses θ with $2\pi\varphi$.

Why the other options are wrong.

(A) $\frac{1}{4}, \frac{3}{4}$ would be $\theta = \frac{\pi}{4}$, i.e. $M/N = \frac{1}{2}$.

(B) $\frac{1}{3}, \frac{2}{3}$ would be $\theta = \frac{\pi}{3}$, i.e. $M/N = \frac{3}{4}$.

(C) This uses $2\pi\varphi = \theta$; the rotation angle is 2θ.

Remember. Quantum counting: $\theta = \pi\varphi$ and $M = N\sin^2\theta$.

Q.68 Single-correct MCQ · Key search

A cipher with a 128-bit key is attacked by Grover's algorithm (one correct key, cost measured in iterations). The number of iterations is about

(A) ✓ $\frac{\pi}{4} \cdot 2^{64}$ **(B)** $\frac{\pi}{4} \cdot 2^{127}$ **(C)** 2^{127} **(D)** 128

Answer: (A) Detailed solution

What you need to know. $k \approx \frac{\pi}{4}\sqrt{N}$ with $N = 2^{128}$. Symmetric keys lose half their bits against Grover.

Step-by-step solution. $N = 2^{128}$, so $k \approx \frac{\pi}{4}\sqrt{N} = \frac{\pi}{4} \cdot 2^{64}$. The classical average is 2^{127}. Doubling the key length restores the original margin. Each iteration also has to run the cipher reversibly, so the real cost is much larger than the iteration count.

Why the other options are wrong.

(B) This uses $\sqrt{N}$ with the wrong exponent.

(C) 2^{127} is the classical average.

(D) 128 is the key length, and search is not polynomial in it.

Remember. Doubling the key length cancels Grover's advantage.

Q.69 Single-correct MCQ · Noise and depth

A simple noise model multiplies the signal by f per iteration: $P_k \approx f^k \sin^2((2k+1)\theta) + (1-f^k)/N$. For $n = 10$, $k = 25$, $f = 0.99$ (take $0.99^{25} \approx 0.78$), the success probability is about

(A) 0.99 **(B)** 0.50 **(C)** ✓ 0.78 **(D)** 0.25

Answer: (C) Detailed solution

What you need to know. The signal decays as f^k, while the background $(1 - f^k)/N$ is very small.

Step-by-step solution. For $n = 10$, $k = 25$ is optimal and $\sin^2(51\theta) \approx 0.9995$, while $(1 - f^k)/N \approx 0.22/1024$ is negligible. So $P \approx 0.78 \times 0.9995 \approx 0.78$. Under this model, a depth of k iterations costs a factor f^k; since $k \propto \sqrt{N}$, this decay is what limits large searches on noisy hardware.

Why the other options are wrong.

(A) 0.99 is the fidelity of one iteration, not of 25.

(B) 0.50 has no basis in the model.

(D) 0.25 confuses $k = 25$ with a probability.

Remember. Long Grover circuits suffer from noise, since $k \propto \sqrt{N}$.

Q.70 Single-correct MCQ · Toffoli count

A multi-controlled Z with m controls can be built with $m-1$ clean ancillas, $2(m-1)$ Toffoli gates and one CZ. For the $n = 10$ Grover register ($m = 9$), one full iteration (oracle and diffusion) needs how many Toffoli gates?

(A) ✓ 32 **(B)** 36 **(C)** 16 **(D)** 18

Answer: (A) Detailed solution

What you need to know. Each iteration has two multi-controlled Z gates. Each costs $2(m-1)$ Toffolis with this construction.

Step-by-step solution. One multi-controlled Z with $m = 9$: $2(9-1) = 16$ Toffolis. An iteration has two of them, one in the oracle and one in the diffusion, so 32 Toffolis. The gate count per iteration is linear in n.

Why the other options are wrong.

(B) $36 = 4m$ also uses m instead of $m - 1$.

(C) 16 counts only one of the two multi-controlled Z gates.

(D) $18 = 2m$ uses the wrong count per gate.

Remember. The cost is linear in n per iteration, and most of it is in the multi-controlled gates.

Q.71 SINGLE-CORRECT MCQ · Simulation cost

On a state-vector simulator the work grows as (iterations) × (amplitudes per gate) $\sim 2^{n/2} \cdot 2^n$. Adding 2 qubits multiplies the total work by about

(A) 16 **(B)** ✓ 8 **(C)** 4 **(D)** $2\sqrt{2}$

Answer: (B) Detailed solution

What you need to know. Work $\sim 2^{1.5n}$: amplitudes double per qubit and iterations grow by $\sqrt{2}$ per qubit.

Step-by-step solution. Work $\propto 2^{1.5n}$, so $n \to n + 2$ gives $2^{1.5\times 2} = 2^3 = 8$: amplitudes grow by 4 and iterations by 2. The number of gates per iteration also grows slowly with n, which this estimate ignores.

Why the other options are wrong.

(A) 16 double-counts: it would need work $\sim 2^{2n}$.

(C) 4 counts only the growth in amplitudes.

(D) $2\sqrt{2}$ is the factor for adding *one* qubit.

Remember. The classical simulation cost is about $N^{1.5}$.

Q.72 SINGLE-CORRECT MCQ · Eigenvectors of G

In the basis $\{|r\rangle, |m\rangle\}$ of the Grover plane, $G = \begin{pmatrix} \cos 2\theta & -\sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{pmatrix}$. Eigenvectors of G are

(A) $\frac{1}{\sqrt{2}}(|r\rangle \pm |m\rangle)$ with eigenvalues $e^{\pm 2i\theta}$

(B) $\frac{1}{\sqrt{2}}(|r\rangle \mp i\,|m\rangle)$ with eigenvalues ± 1

(C) ✓ $\frac{1}{\sqrt{2}}(|r\rangle \mp i\,|m\rangle)$ with eigenvalues $e^{\pm 2i\theta}$

(D) $|r\rangle$ and $|m\rangle$ with eigenvalues $e^{\pm 2i\theta}$

Answer: (C) Detailed solution

What you need to know. A plane rotation has the complex eigenvectors $(1, \mp i)^T/\sqrt{2}$.

Step-by-step solution. Apply G to $(1, -i)^T$: the first entry is $\cos 2\theta + i \sin 2\theta = e^{2i\theta}$; the second is $\sin 2\theta - i \cos 2\theta = -i\, e^{2i\theta}$. So $(1, -i)^T$ has eigenvalue $e^{+2i\theta}$, and by conjugation $(1, +i)^T$ has $e^{-2i\theta}$. These eigenvectors are complex: a rotation by an angle other than 0 or π has no real eigenvector.

Why the other options are wrong.

(A) Real vectors are not eigenvectors of a rotation except for angles 0 and π.

(B) The vectors are right but ± 1 are not the eigenvalues; $|\lambda| = 1$ alone does not make them real.

(D) $G\,|r\rangle$ has components on both axes, so $|r\rangle$ is not an eigenvector.

Remember. $|s\rangle$ is an equal-weight superposition of the two eigenvectors, which is why counting works.

Q.73 Single-correct MCQ · Fixed-point search

In Grover's fixed-point ($\pi/3$) search, one recursion step turns a procedure with failure probability ε into one with failure probability ε^3. Starting from $\varepsilon = 0.1$, two steps give

(A) 10^{-3} **(B)** 10^{-6} **(C)** 10^{-27} **(D)** ✓ 10^{-9}

Answer: (D) Detailed solution

What you need to know. Recursion: $\varepsilon \mapsto \varepsilon^3$, and thus $\varepsilon_j = \varepsilon^{3^j}$.
Step-by-step solution. After one step $\varepsilon_1 = (0.1)^3 = 10^{-3}$; after the second $\varepsilon_2 = (10^{-3})^3 = 10^{-9}$. In general $\varepsilon_j = \varepsilon^{3^j}$. Unlike ordinary Grover, additional steps never reduce the success probability, so the method cannot over-rotate; the price is that it loses the quadratic speed-up in this basic form.
Why the other options are wrong.
(A) 10^{-3} is the result of one step.
(B) 10^{-6} treats the second step as a multiplication by 10^{-3}.
(C) 10^{-27} is the result after three steps.
Remember. Fixed-point search trades speed for safety against over-rotation.

Q.74 Single-correct MCQ · Repeat-until-success cost

Let $N = 8$, $M = 1$, and repeat a run with a fixed k until the marked item is found, counting only oracle queries in the Grover iterations. Then $P_1 = \frac{25}{32}$ and $P_2 = \frac{121}{128}$. The expected numbers of queries for $k = 1$ and $k = 2$ are about

(A) 1 and 2 **(B)** 2.12 and 1.28 **(C)** ✓ 1.28 and 2.12 **(D)** 0.78 and 1.89

Answer: (C) Detailed solution

What you need to know. Expected cost of repeat-until-success is k/P_k.
Step-by-step solution. The number of runs is geometric with mean $1/P_k$, and each run costs k queries, so the expected cost is k/P_k. $k = 1$: $\frac{32}{25} = 1.28$. $k = 2$: $2 \times \frac{128}{121} \approx 2.12$. So $k = 1$ is cheaper here, although $k = 2$ is the value that maximises the success probability of a single run: the two criteria are different.
Why the other options are wrong.
(A) Those are k, which ignores repetition after failed runs.
(B) The values are in the wrong order.
(D) These are P_1 and kP_2 rather than k/P_k.
Remember. The k that maximises P_k need not minimise the expected cost.

Q.75 Matching lists · Query complexities

Match each task in List-I with its query complexity in List-II.

List-I		List-II	
(P)	Unstructured search, classical	**(1)**	$\Theta(N^{1/3})$
(Q)	Unstructured search, quantum	**(2)**	$\Theta(N^{2/3})$
(R)	Collision finding in a 2-to-1 function, quantum	**(3)**	$\Theta(N)$
(S)	Element distinctness, quantum	**(4)**	$\Theta(N^{1/2})$

(A) ✓ P→3, Q→4, R→1, S→2 **(B)** P→4, Q→3, R→1, S→2
(C) P→3, Q→1, R→4, S→2 **(D)** P→3, Q→4, R→2, S→1

Answer: (A) Detailed solution

What you need to know. Known query complexities: search N vs. $\sqrt{N}$, collision $N^{1/3}$, element distinctness $N^{2/3}$.

Step-by-step solution. Classical search must examine a constant fraction of the items: $\Theta(N)$ (3). Grover: $\Theta(N^{1/2})$ (4). Brassard–Høyer–Tapp collision finding, optimal by Aaronson–Shi: $\Theta(N^{1/3})$ (1). Ambainis's quantum walk for element distinctness: $\Theta(N^{2/3})$ (2). Hence P→3, Q→4, R→1, S→2.

Why the other options are wrong.

(B) This exchanges classical and quantum search.

(C) This assigns $N^{1/3}$ to unstructured search, which contradicts the lower bound.

(D) This exchanges collision and element distinctness; the promise of 2-to-1 makes collision easier.

Remember. More structure leads to lower query complexity.

Q.76 MATCHING LISTS · Variants of Grover

Match each method in List-I with its main idea in List-II.

List-I		List-II	
(P)	Amplitude amplification	**(1)**	uses randomly chosen k from a growing range when M is unknown
(Q)	Quantum counting	**(2)**	changes the reflection phases so that extra steps cannot over-rotate
(R)	BBHT search	**(3)**	replaces $H^{\otimes n}$ by any state-preparation procedure $\mathcal{A}$
(S)	Fixed-point search	**(4)**	applies phase estimation to G to find M

(A) P→3, Q→1, R→4, S→2

(B) ✓ P→3, Q→4, R→1, S→2

(C) P→3, Q→4, R→2, S→1

(D) P→4, Q→3, R→1, S→2

Answer: (B) Detailed solution

What you need to know. Each variant answers one weakness: arbitrary start, unknown M, or over-rotation.

Step-by-step solution. Amplitude amplification generalises $|s\rangle$ to $\mathcal{A}|0\rangle$ (3). Quantum counting reads θ from the eigenvalues $e^{\pm 2i\theta}$ of G, which yields M (4). BBHT draws k at random from a range that grows geometrically (1). Fixed-point search uses reflections with phase $\pi/3$ and similar choices rather than π (2). Hence P→3, Q→4, R→1, S→2.

Why the other options are wrong.

(A) This exchanges Q and R.

(C) This exchanges R and S; fixed-point search changes the phases and BBHT randomises k.

(D) This exchanges P and Q; quantum counting uses phase estimation.

Remember. Learn the problem each variant solves, not only its name.

Q.77 Assertion–Reason · Grover and NP-completeness

Assertion (A): Grover's algorithm solves NP-complete problems in polynomial time.

Reason (R): For a search over 2^n candidates, Grover's algorithm uses about $\frac{\pi}{4}2^{n/2}$ iterations.

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A

(B) Both A and R are true, but R is *not* the correct explanation of A

(C) A is true, but R is false

(D) ✓ A is false, but R is true

Answer: (D) Detailed solution

What you need to know. Quadratic speed-up does not make an exponential polynomial.
Step-by-step solution. **R is true:** this is $\frac{\pi}{4}\sqrt{N}$ with $N = 2^n$. **A is false:** $2^{n/2}$ is still exponential in n. The speed-up is quadratic, and the black-box lower bound shows that generic search cannot do better. Answer: (D).
Why the other options are wrong.

(A) A is false, so this option fails.

(B) A is false, so this option fails even though R is true.

(C) A is false and R is true, the opposite of this option.

Remember. $2^{n/2}$ is still exponential.

Q.78 Statement I / II · Unknown M; flat curve at $M = N/2$

Statement I: The BBHT algorithm finds a marked item with an expected $O(\sqrt{N/M})$ queries without knowing M in advance.

Statement II: If exactly half of the items are marked, ordinary Grover iterations leave the success probability at $\frac{1}{2}$.

(A) ✓ Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true, but Statement II is false

(D) Statement I is false, but Statement II is true

Answer: (A) Detailed solution

What you need to know. BBHT handles unknown M; and $\theta = 45°$ gives a flat success curve.
Step-by-step solution. **I is true:** a growing range for the random k reaches $m \sim \sqrt{N/M}$ after a geometric series of attempts, whose total cost is dominated by the final terms. **II is true:** $\theta = 45°$ and $\sin^2((2k+1)45°) = \frac{1}{2}$ for all k (see Q.50). Answer: (A).
Why the other options are wrong.

(B) Both statements are true, not false.

(C) II is also true, as $\sin^2$ of odd multiples of 45° is $\frac{1}{2}$.

(D) I is also true: the expected cost is $O(\sqrt{N/M})$.

Remember. Unknown M is solved, and $M = N/2$ is not a problem since simply measuring works.

Paragraph for Questions 79 and 80

Quantum counting. A search space of $N = 64$ items contains an unknown number M of marked items; in fact $M = 16$. The Grover operator G has eigenvalues $e^{\pm 2i\theta}$ on the plane containing $|s\rangle$, with $\sin^2\theta = M/N$. Phase estimation of G with $t = 3$ counting qubits is run on $|s\rangle$, which has equal weight on the two eigenvectors. The outcome $y \in \{0, \ldots, 7\}$ gives the phase estimate $\varphi \approx y/8$, where $e^{2\pi i\varphi}$ is the eigenvalue. From this one sets $\tilde{\theta} = \pi y/8$ and $\tilde{M} = N\sin^2\tilde{\theta}$.

Q.79 PARAGRAPH BASED · Quantum counting outcomes

The two most likely outcomes are

(A) $y = 2$ and $y = 6$ **(B)** ✓ $y = 1$ and $y = 7$ **(C)** $y = 1$ and $y = 2$ **(D)** $y = 0$ and $y = 4$

Answer: (B) Detailed solution

What you need to know. $\varphi = \theta/\pi$ and $1 - \theta/\pi$. QPE peaks at the grid point nearest to $2^t\varphi$.
Step-by-step solution. $\theta = \arcsin\sqrt{16/64} = \arcsin\frac{1}{2} = \frac{\pi}{6}$, so the eigenphases are $\varphi = \frac{\theta}{\pi} = \frac{1}{6}$ and $\frac{5}{6}$. On the grid $y/8$ these lie at $y = \frac{8}{6} = 1.33$ and $y = \frac{40}{6} = 6.67$; the closest integers are 1 and 7. Each has probability about 0.35. (The next most likely are $y = 2$ and $y = 6$, each about 0.09.)
Why the other options are wrong.
(A) $y = 2, 6$ are the next most likely, each with about 0.09.
(C) Both eigenphases contribute, and they are symmetric: y and $8 - y$.
(D) $y = 0, 4$ would be $\theta = 0$ and $\theta = \frac{\pi}{2}$.
Remember. Outcomes come in pairs y and $2^t - y$, which give the same $\sin^2\tilde{\theta}$.

Q.80 PARAGRAPH BASED · Quantum counting estimate

If $y = 1$ is obtained, the estimate of M is

(A) $\tilde{M} = 32$ **(B)** $\tilde{M} = 8$ **(C)** ✓ $\tilde{M} \approx 9.4$ **(D)** $\tilde{M} = 16$

Answer: (C) Detailed solution

What you need to know. $\tilde{M} = N\sin^2(\pi y/2^t)$; $\sin^2\frac{\pi}{8} = \frac{1-\cos(\pi/4)}{2}$.
Step-by-step solution. $\tilde{\theta} = \frac{\pi}{8}$ and $\tilde{M} = 64\sin^2\frac{\pi}{8} = 64 \cdot \frac{1-\cos(\pi/4)}{2} = 32(1 - \frac{\sqrt{2}}{2}) \approx 9.37$. The true value 16 lies between the estimates from $y = 1$ (9.4) and $y = 2$ (32): three counting qubits are too few for an accurate count. Each additional counting qubit halves the error in θ.
Why the other options are wrong.
(A) 32 is the estimate from $y = 2$.
(B) 8 uses $\sin^2\frac{\pi}{8} \approx \frac{1}{8}$ and is not the correct value.
(D) 16 is the true value, but $y = 1$ gives a coarse estimate.
Remember. More counting qubits give a finer grid and a better estimate of M.